

DÆDÆLUS

Unbreakable Networks For Distributed Applications

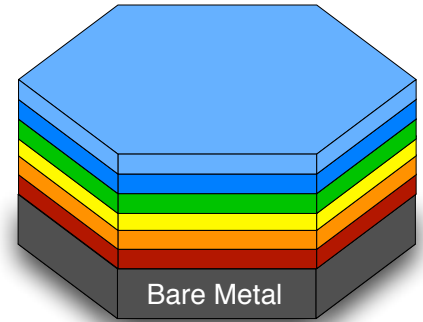
Animated Slides

From Archive: August 2nd, 2019

This is completely new

Misconceptions to Avoid

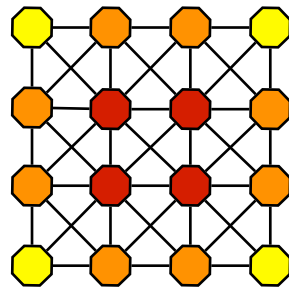
- ▶ This will not fit one of your pigeon holes
- ▶ This is not a science experiment
- ▶ This is not boiling the ocean
- ▶ We subsume legacy (but not bogged down by it)
- ▶ There's something deeper going on here than it looks
- ▶ There's something much simpler going on here than it appears
- ▶ There's something profoundly more resilient than thought possible



Spoilers

Purpose-built Cellular Fabrix for distributed applications

- ▶ **New addressing mechanism** (Hijack standard Ethernet frames)
 - ▶ **R**eliable **A**ddress **F**ree **E**thernet
- ▶ **Highly resilient, partition-free fabric** (High NIC's)
 - ▶ Intrinsic failure detection and Exactly-Once delivery
- ▶ **Secret Sauce: Atomic State Transfer (AIT)**
 - ▶ Microtransactions for low latency
 - ▶ Consensus Tiles for high resilience



Chiplet Fabrix

Summary: Fast Secure Transactions on Graphs

- ▶ New addressing mechanism (Hijack standard Ethernet frames)

- ▶ **S**ecure **A**ddress **F**ree **E**thernet

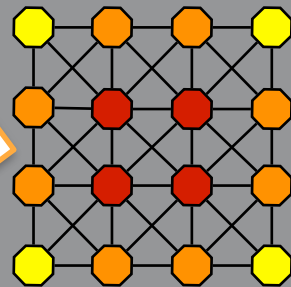
- ▶ Highly resilient, partition-free fabric through NIC

- ▶ Intrinsic failure detection and Exactly-Once delivery

- ▶ Secret Sauce: Atomic Information Transactions (AIT)

- ▶ Micro-transactions for low latency

- ▶ Consensus Tiles for high performance



Secure: Direct Fabrix

Graph
Infrastructure
Programming

Simple

Resilient

Secure

Scalable

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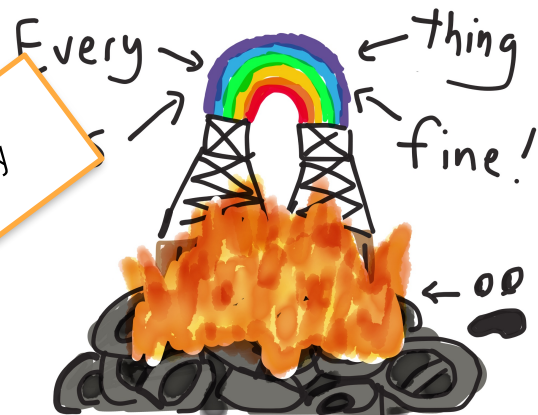
Ethernet Networks Today

Complex, Unreliable, Vulnerable

Drop, Reorder, Duplicate & Delay packets:

- ▶ Cascade Failures
- ▶ Transaction Failures
- ▶ Consistency Failures
- ▶ Impossible to achieve "Exactly Once Semantics"

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[Jepsen 9: A Fsyncing Feeling](#)
[Kyle Kingsbury/GOTO Conference April 2018](#)

All mitigations are band-aids – and are never enough

Problem:

Networks drop,
delay, duplicate and
reorder packets

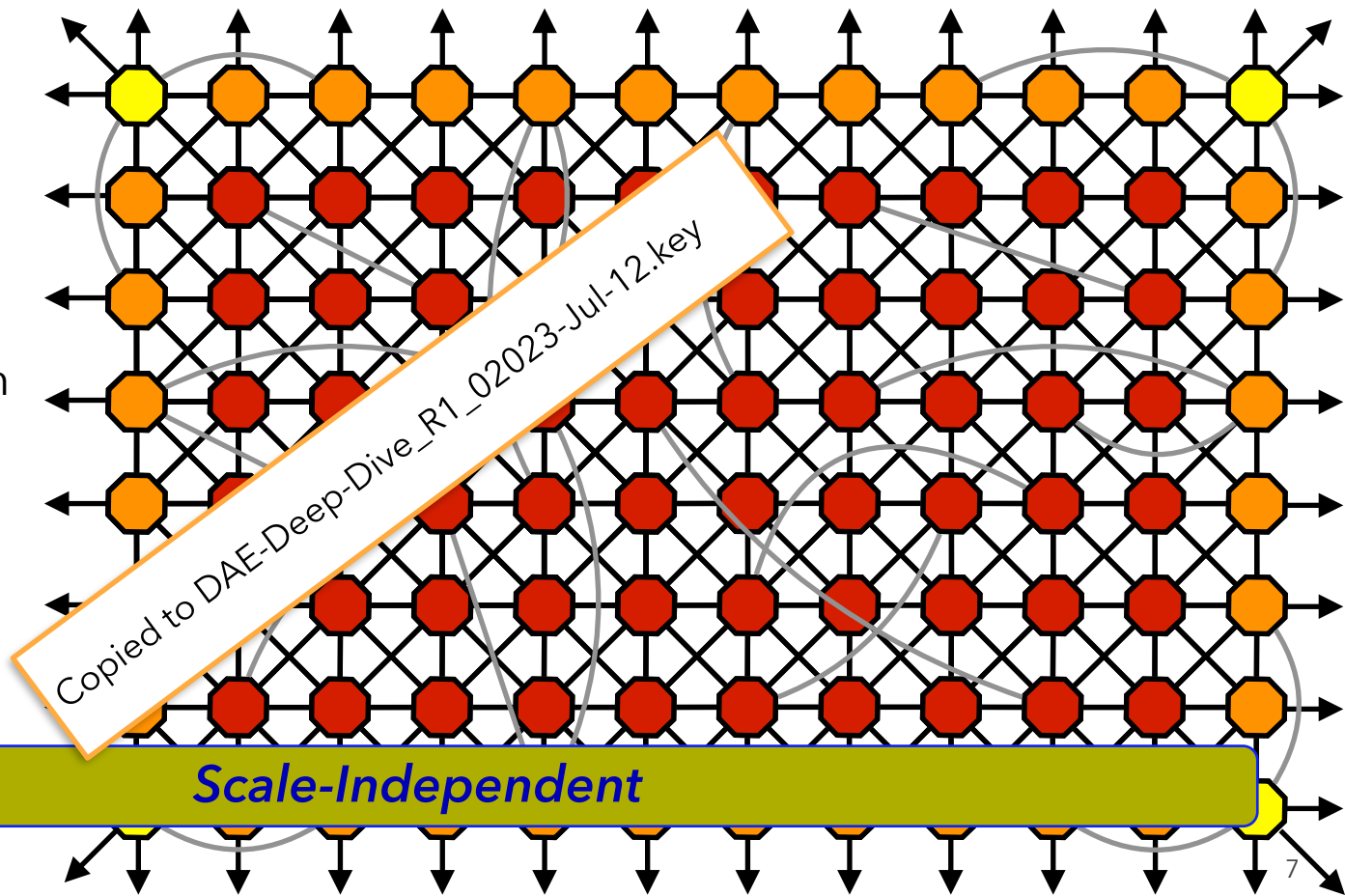
Ruins consistency.
Precipitates failure
cascades

The industry is
missing something
critically important



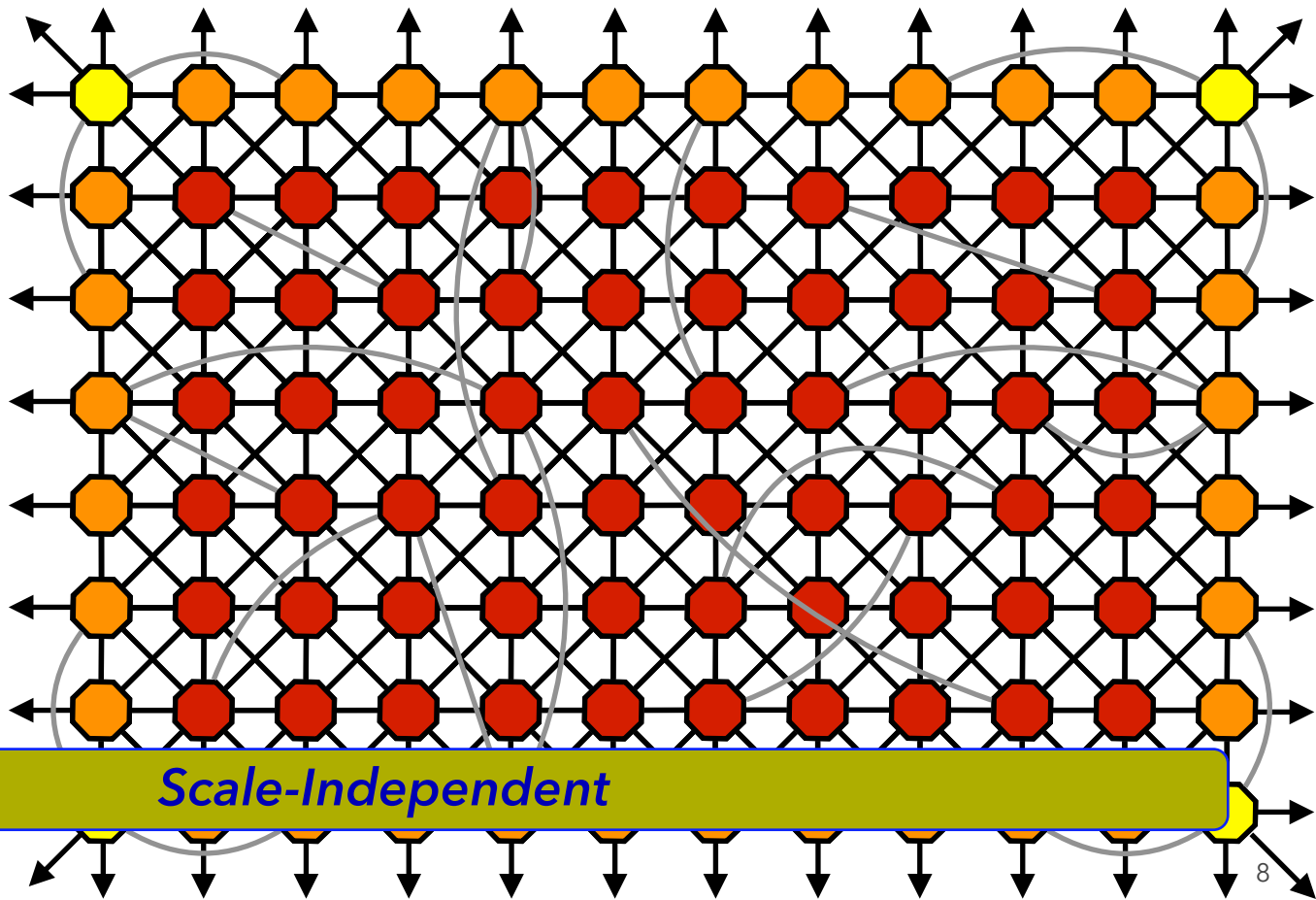
Cellular Fabrix

- ▶ Start with a Cell
- ▶ Connect only to its neighbors
- ▶ Linearly scale them
- ▶ Toss in a few Kleinberg Links to reduce average hops



Cellular Fabrix

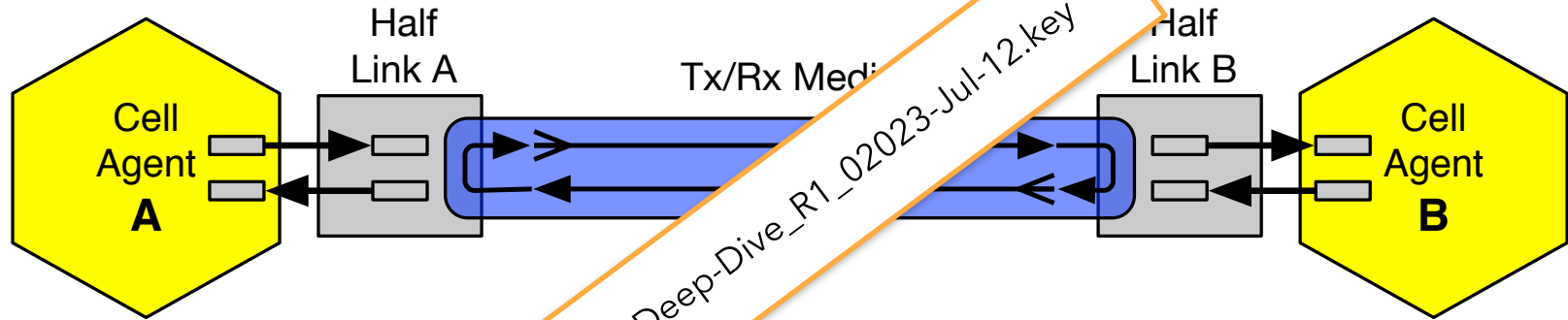
- ▶ Start with a Cell
- ▶ Connect only to its neighbors
- ▶ Linearly scale them
- ▶ Toss in a few Kleinberg Links to reduce average hops



Secret Sauce: Active Link Technology

Atomic Information Transfer (AIT)

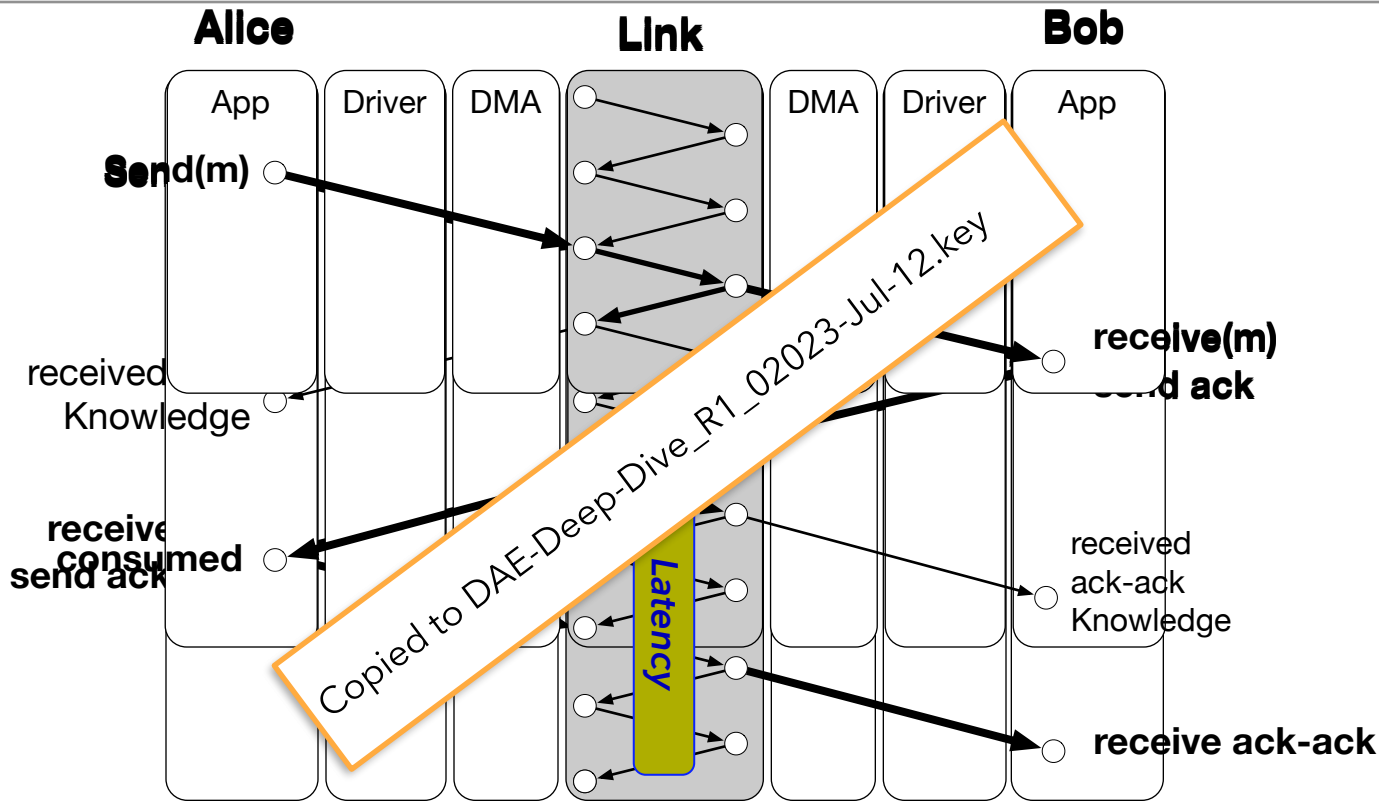
Guaranteed in the sense that both sides "know" if it failed or succeeded



- A new protocol over any physical link
- Reversible
- No-Copying
- First class object
- Composable

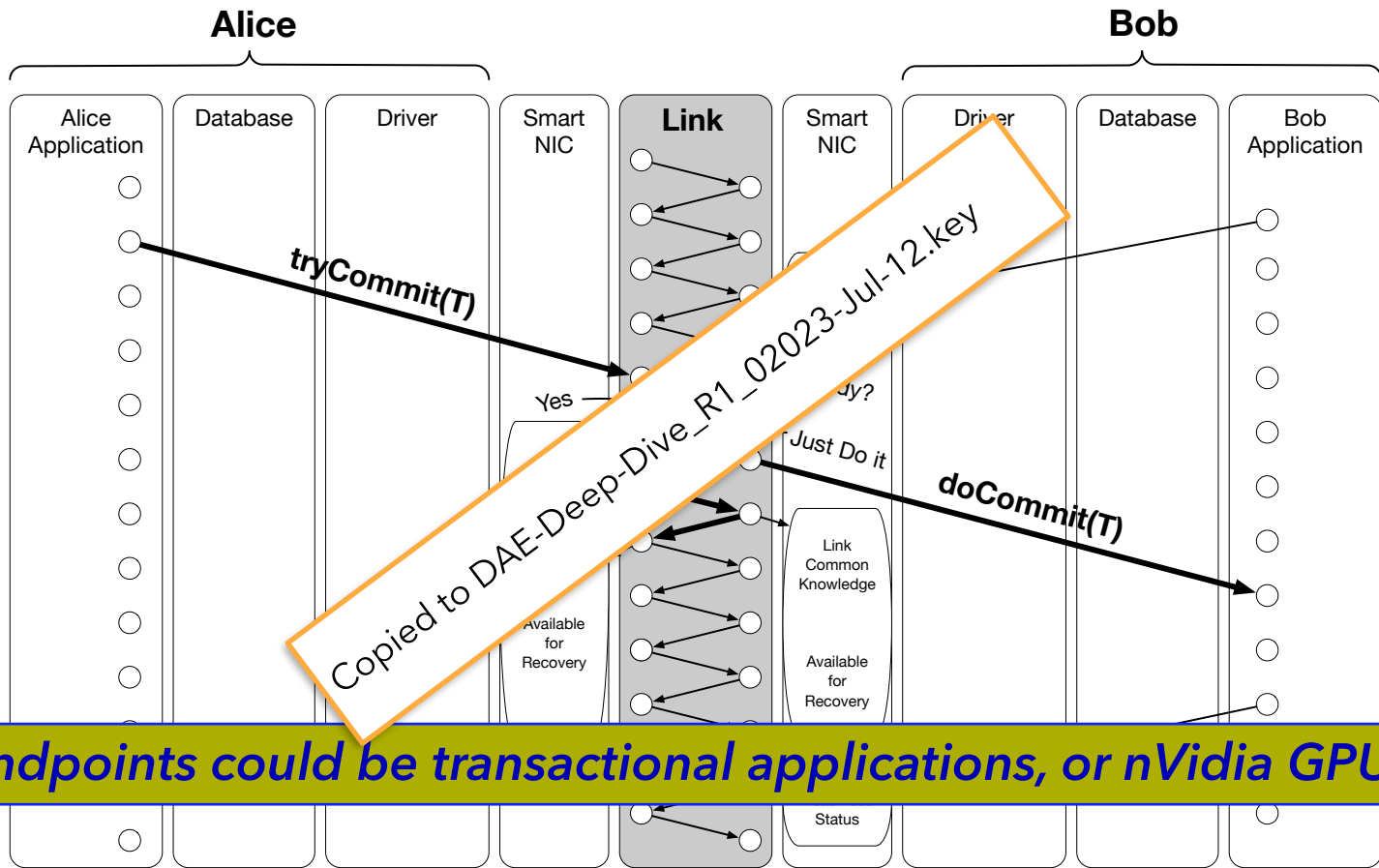
Information can't travel faster than light, but liveness can

Performance Benefits



Reliability & Performance Using Common Knowledge in the Link

Implications



GVM

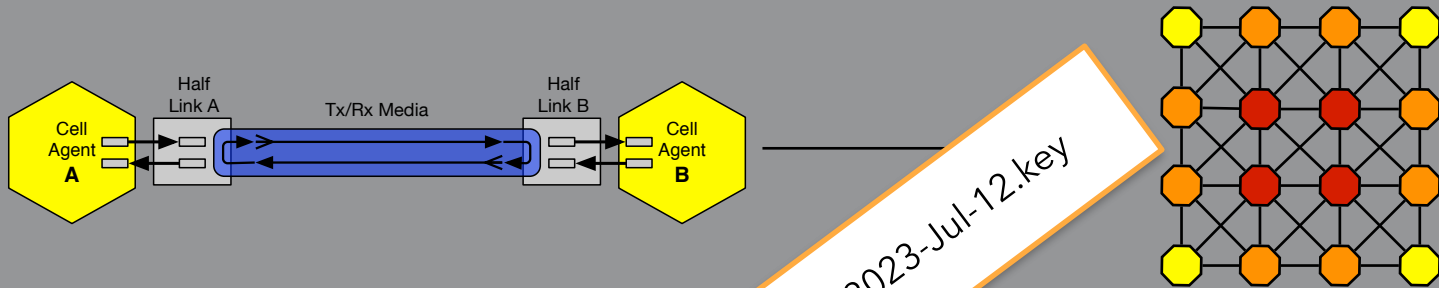
Graph Virtual Machine

Orchestration for

secure

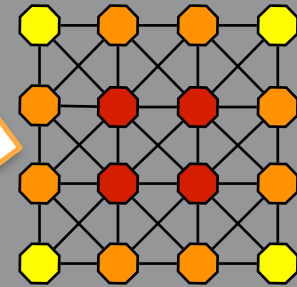
operations on graphs

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Fast: Low Latency Link

Secure: Cellular Fabrix



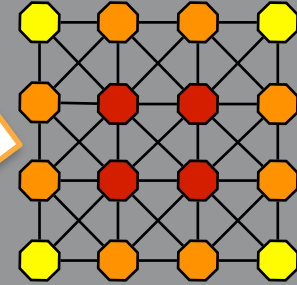
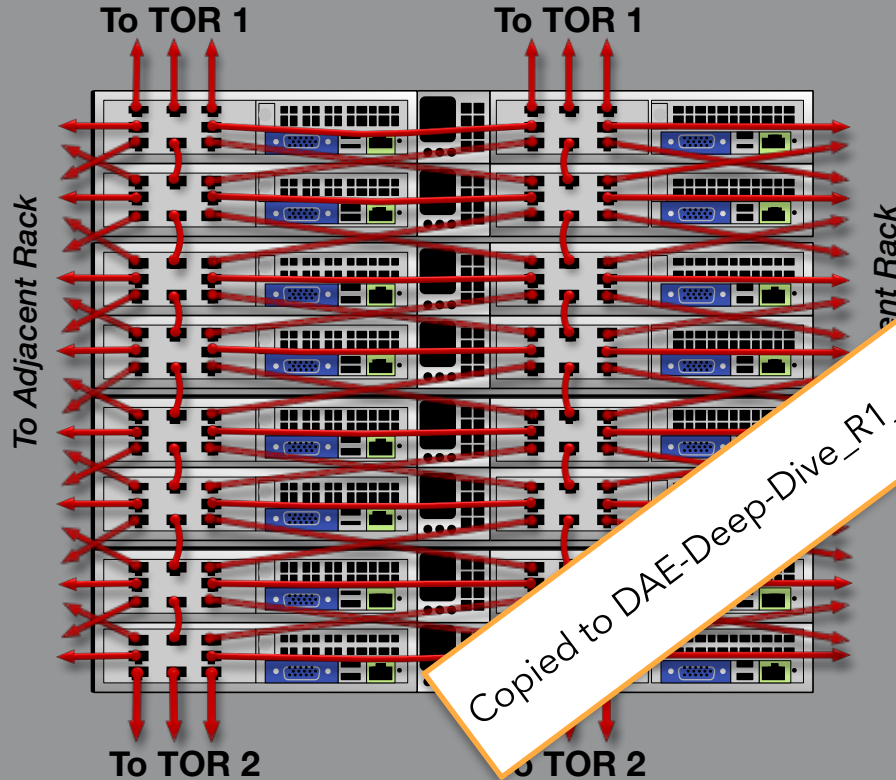
transacti

on

graphs

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Zero Touch (no intervention required)

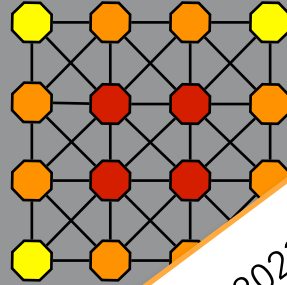


Secure: Direct Fabrix

graphs

Automatic Discovery, Configuration & Adapts to Cabling Errors

Zero Latency



Memory Centric (MM) Clusters

Eliminate Latency by Bypassing Switches

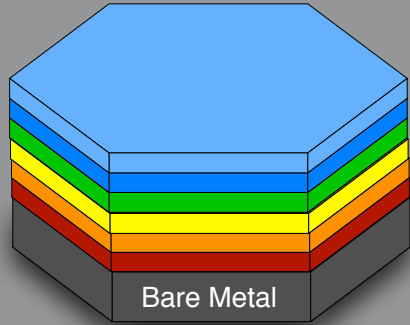
Eliminate I/O Bottlenecks with Quantum-like Protocols

Eliminate Protocol Stacks, Filesystems, DRAM Double Access Penalty from Storage

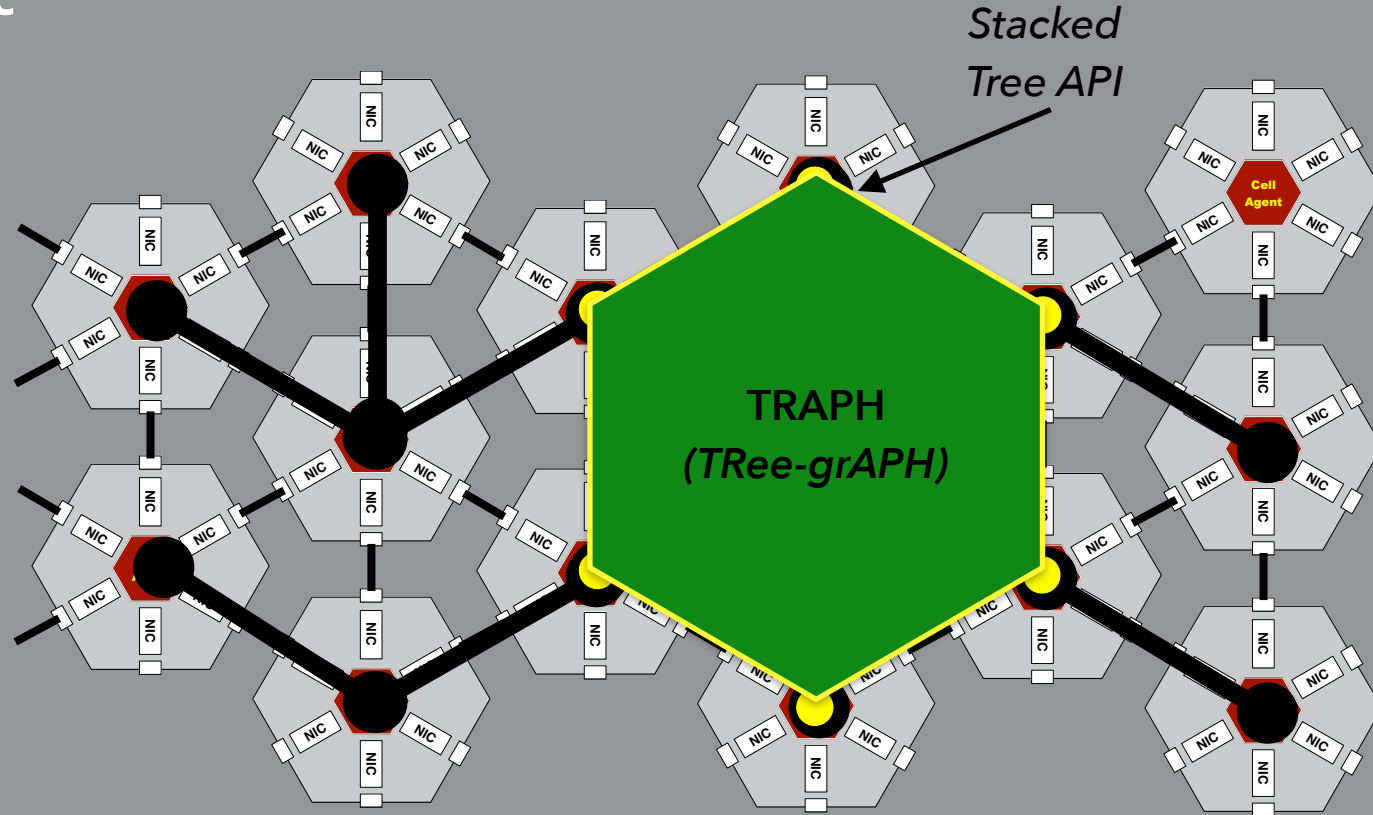
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Zero Trust

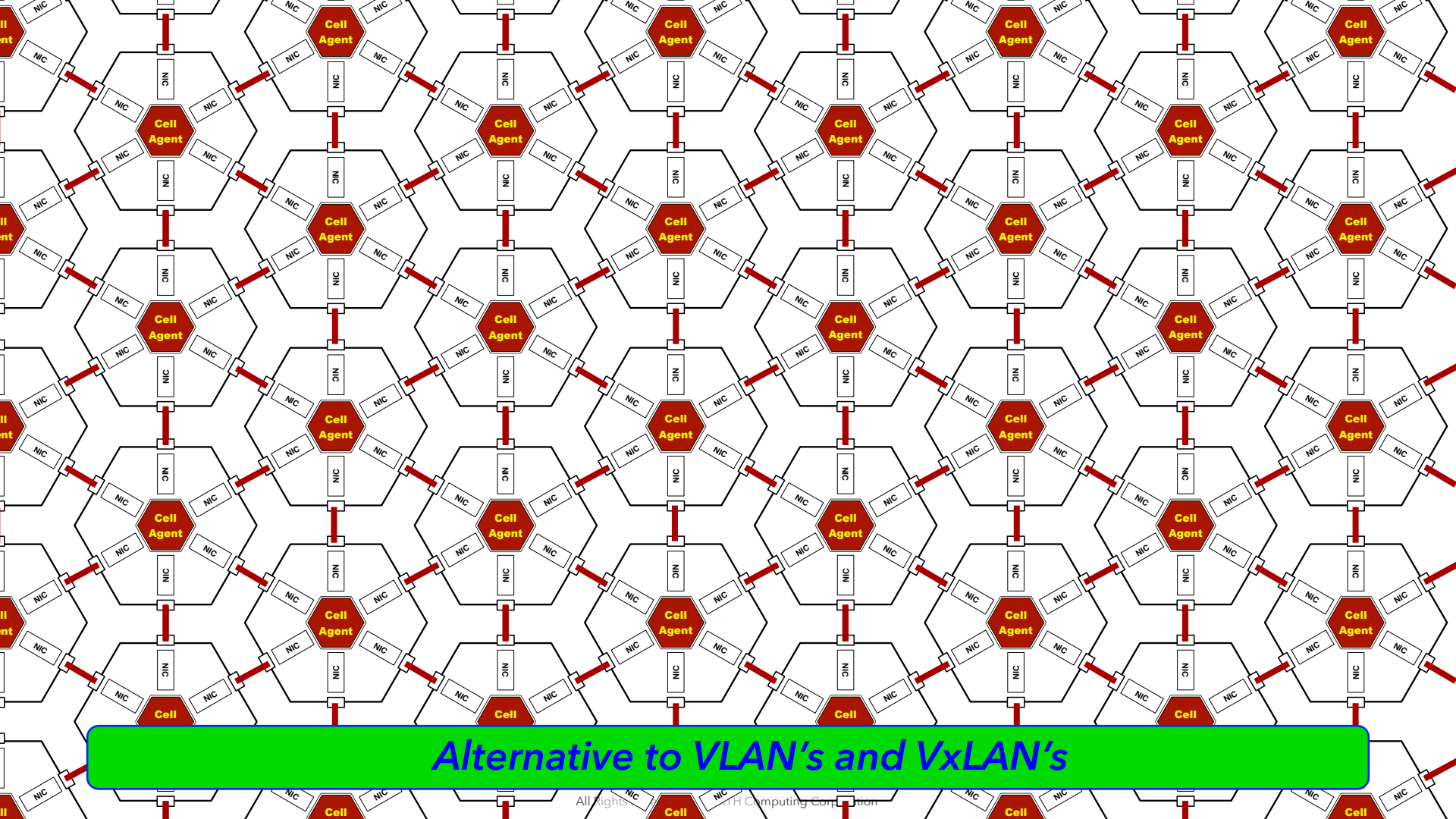
Legacy



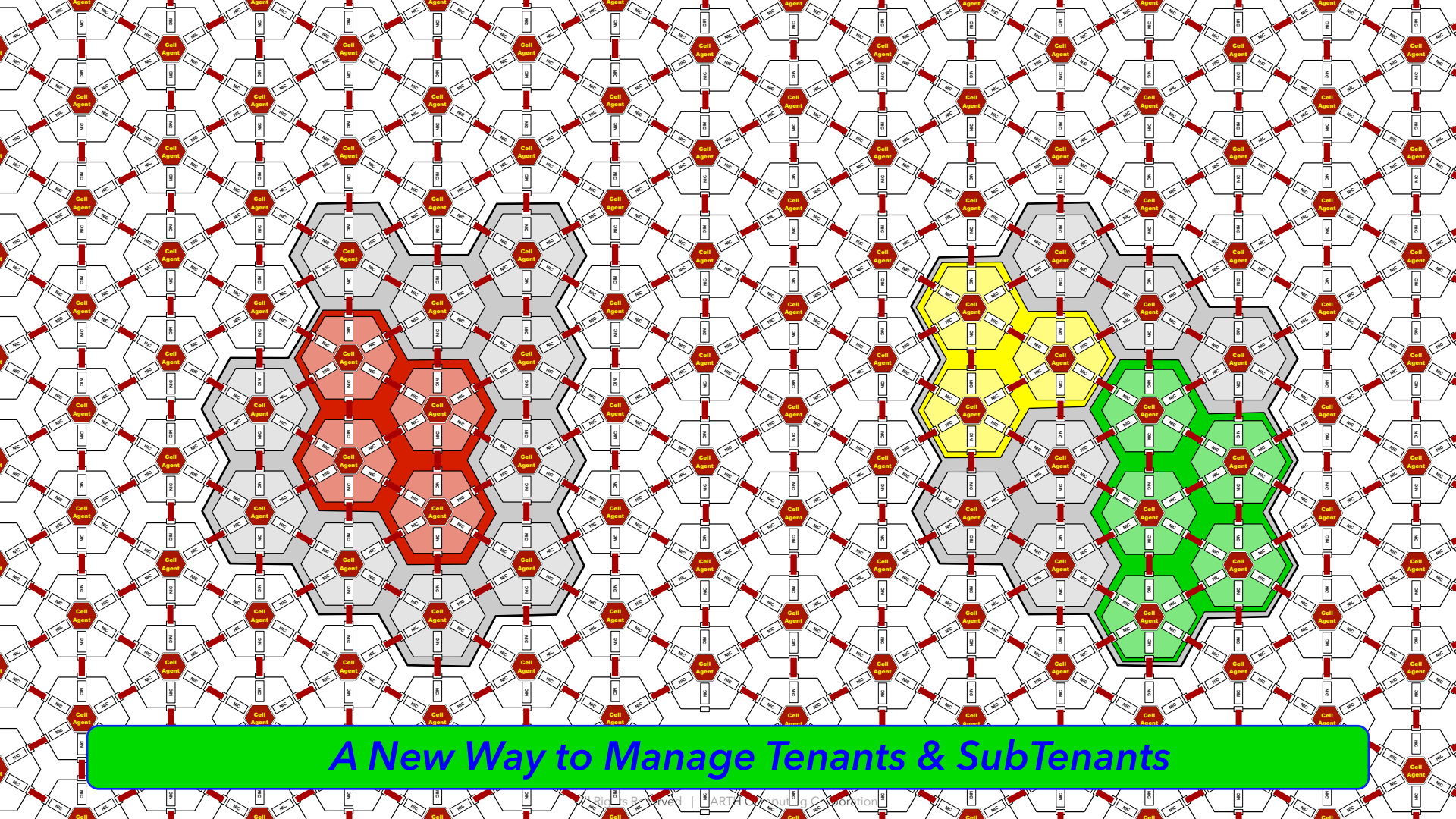
Commodity



Infrastructure Slices based on Stacked Trees



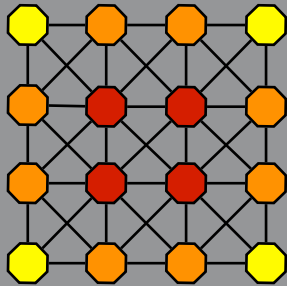
Alternative to VLAN's and VxLAN's



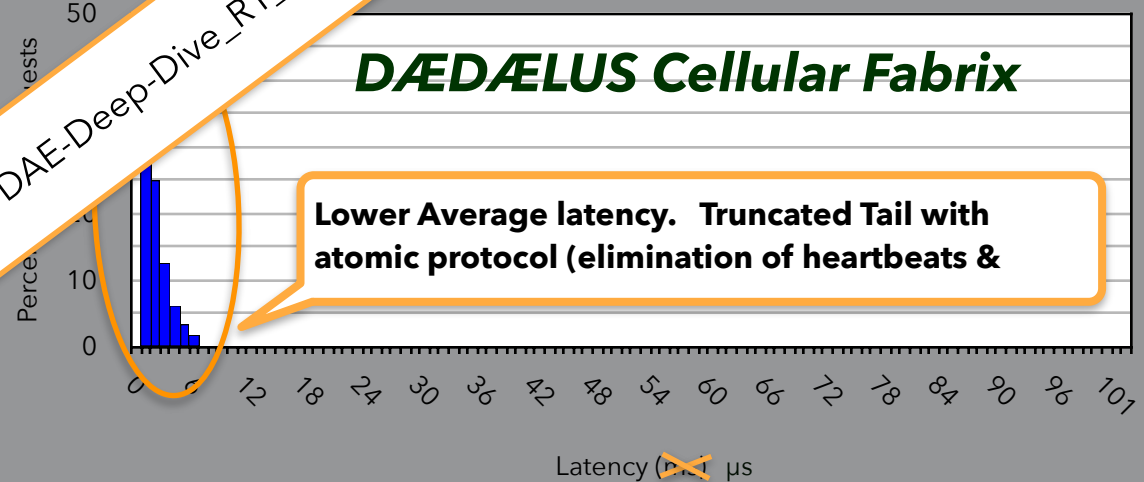
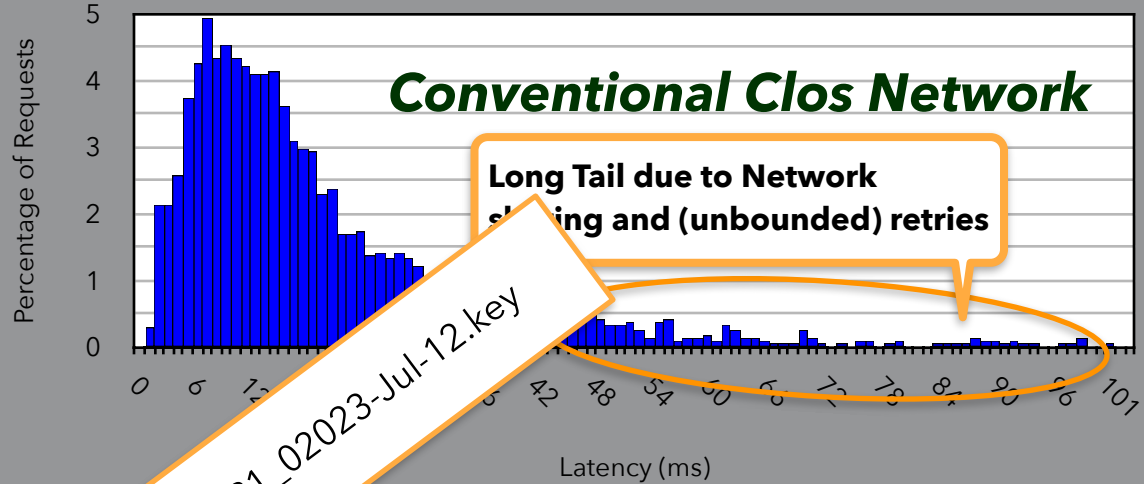
A New Way to Manage Tenants & SubTenants

Latency

- Cellular Fabrix reduces latency in ways conventional networks cannot:
 - Direct connections
 - Multicast requests (e.g. consensus) spread over 8+ ports instead of 1
 - Truncated Tail Latency – protocol knows it failed or succeeded (without heartbeats or timeouts)



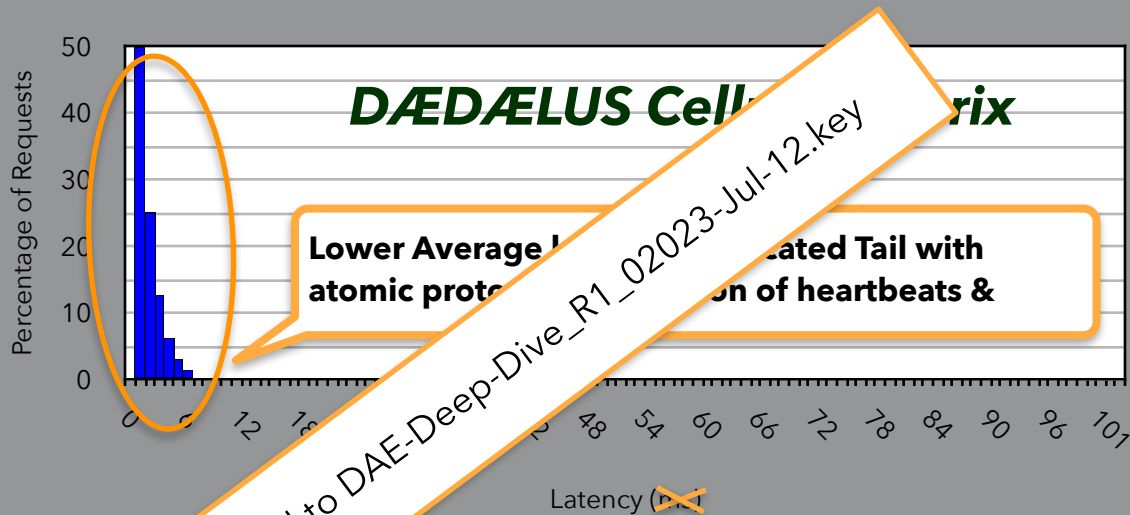
Cellular Fabrix



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Latency: There's plenty of room at the bottom

1 microsecond is 1/3 Kilometer – 1 millisecond is 300 Km (186 miles)



microseconds

nanoseconds



Bob



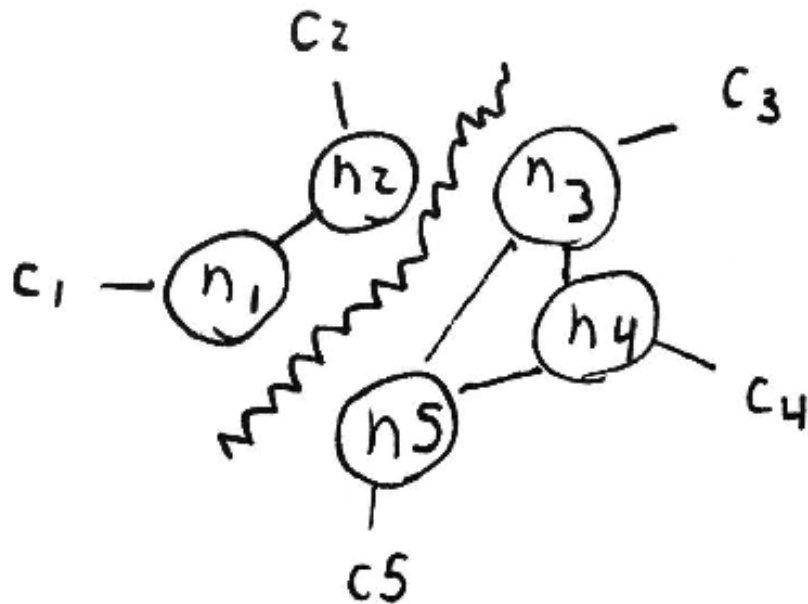
Alice

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Demo

ed Token Coherence

Dramatic Benefits: Distributed Systems



Effective Elimination of Partitions

- *No CAP Theorem Tradeoffs*
- *No FLP Recovery*

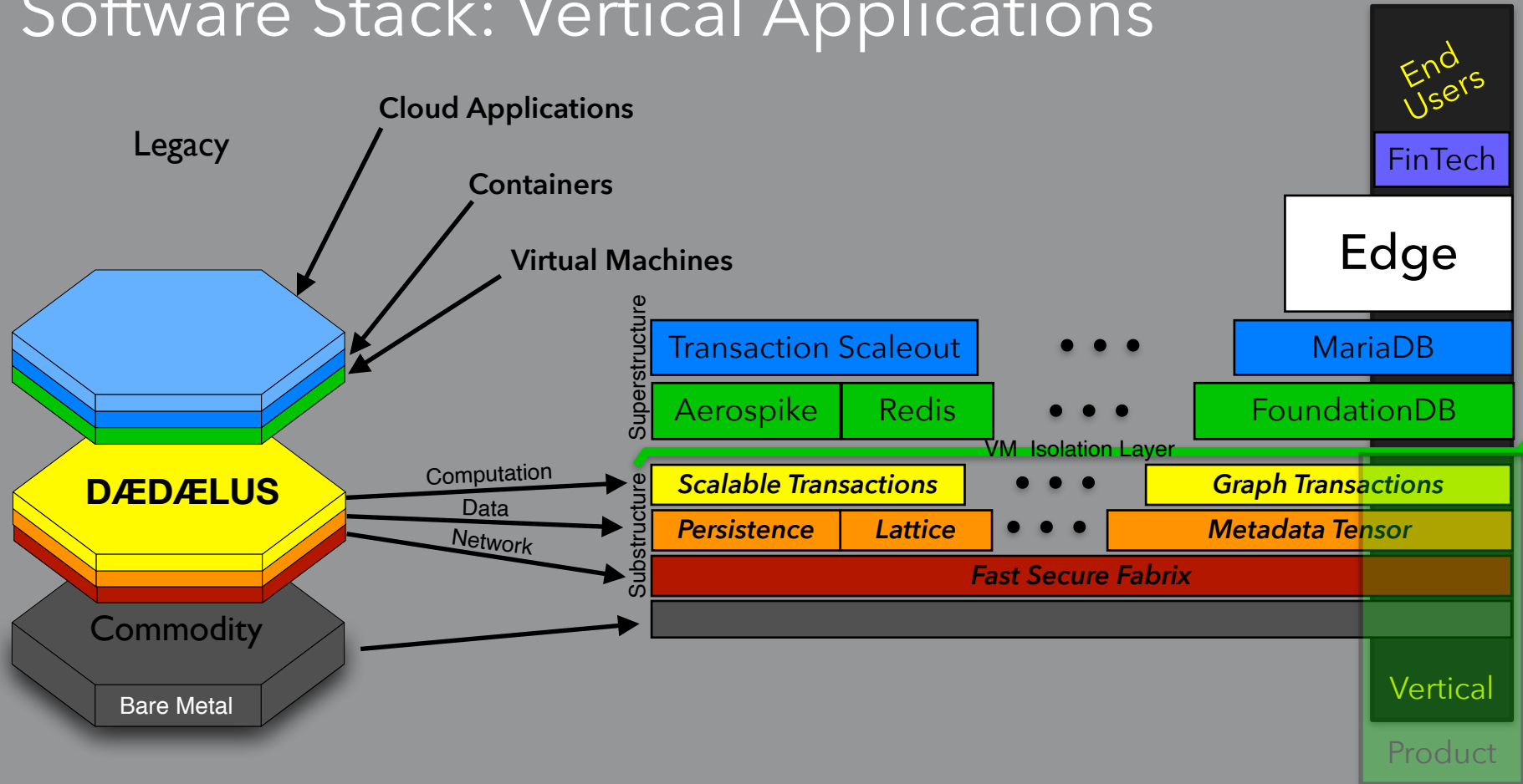
No Timeout Storms

No Reconstruction Storms

Simple Load Balancing

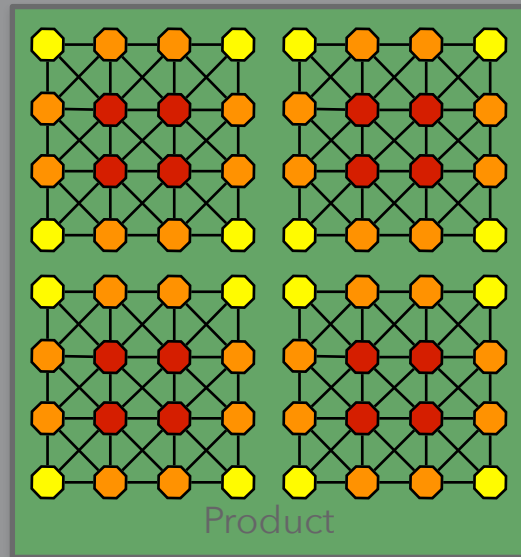
Simple Foundation for Distributed Systems

Software Stack: Vertical Applications



Market Entry: Geographically Dispersed Datacenters

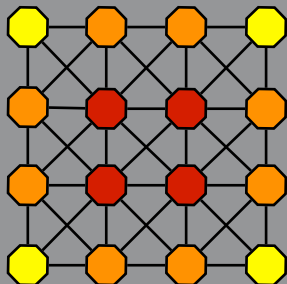
- ▶ Partner with Bare Metal Cloud providers, e.g.:
[Packet](#), [RackSpace](#), [Digital Ocean](#), [Fastly](#), [Nutanix](#)
 - ▶ Port their Apps to our fabric, one at a time
- ▶ Credit card: \$99/month per consensus tile (on top standard rates for bare metal cloud provider).
 - ▶ Competitive with or half AWS prices
 - ▶ Offer 3rd-party managed services above green line
- ▶ Work closely with edge providers – AT&T, Verizon & Comcast for their microdatacenters
- ▶ Ultimate charging model:
 - ▶ The resolution of uncertainty



Stand up
MicroDatacenters
One Rack at a Time

Cellular Fabrix Benefits

No SPOF's
No Partitions
No Packet Loss
No Reordering
Exactly Once



Resilience

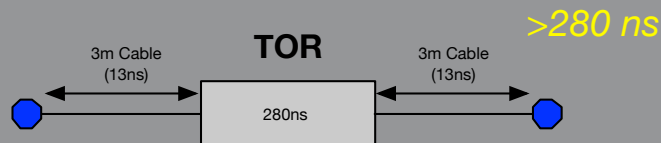
Short cables

No switch delays
(or packet losses)
within a rack

Cellular Fabric

<1 ns

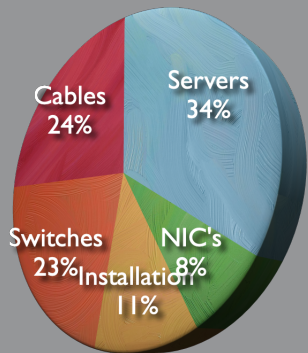
0.3m Cable (2ns)



>280 ns

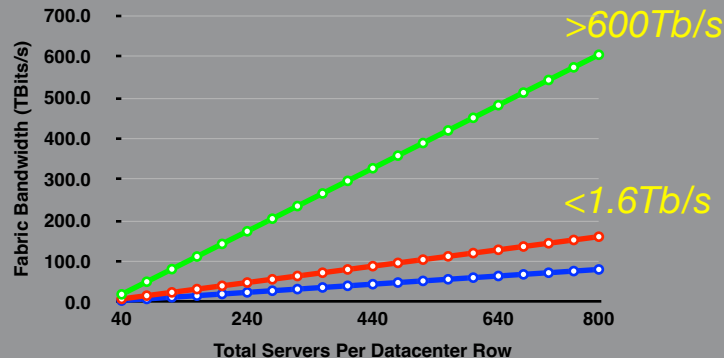
Latency

If switches were
free, Clos would
still be more
expensive



Cost

Ten Times the
fabric
bandwidth at
half the cost



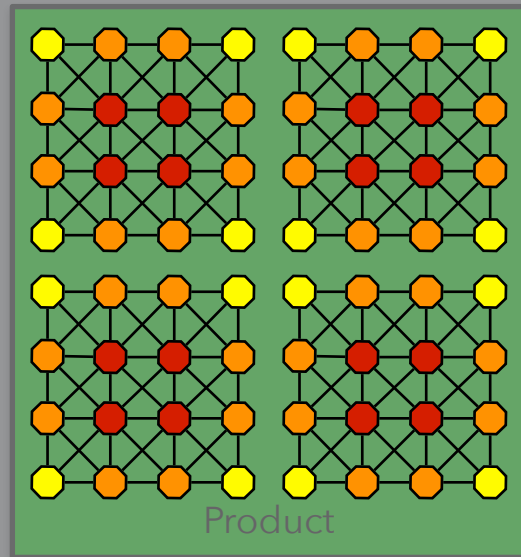
>600Tb/s

<1.6Tb/s

Bandwidth

Market Entry: Chiplet MicroDatacenters

- ▶ Partner with Cohosting Cloud providers, e.g.:
[Packet](#), [Switch](#), [Digital Ocean](#), [Fastly](#), [Equinix](#)
 - ▶ Port their Apps to our fabric, one at a time
- ▶ Work closely with edge providers – AT&T, Verizon & Comcast for their microdatacenters
 - ▶ Provision Micro-slices of infrastructure based on our Stacked Tree technology



Stand up
ChipletDatacenters
One Rack at a Time

Summary: Fast Secure Transactions on Graphs

- New addressing mechanism (Hijack standard Ethernet frames)

- **R**eliable **A**ddress **F**ree **E**thernet

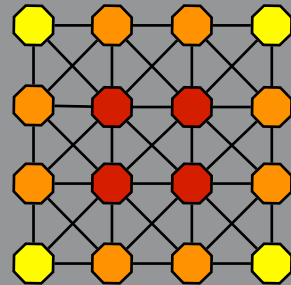
- Highly resilient, partition-free fabric through NIC's

- Intrinsic failure detection and Exactly-Once delivery

- Secret Sauce: Atomic Information Transfer (AIT)

- Microtransactions for low latency

- Consensus Tiles for high resilience



Secure: Direct Fabric

Graph
Infrastructure
Programming

Simple

Resilient

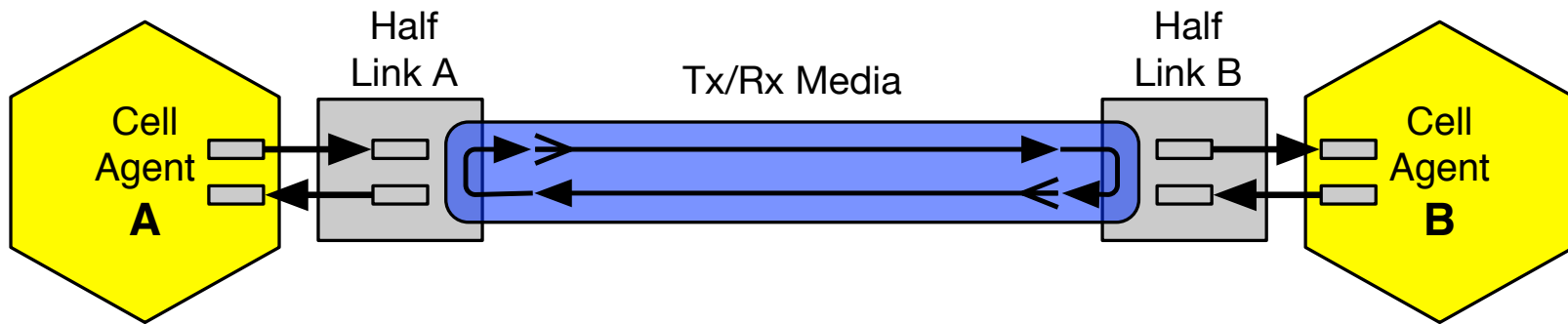
Secure

Scalable

Secret Sauce: Active Link Technology

Atomic Information Transfer (AIT)

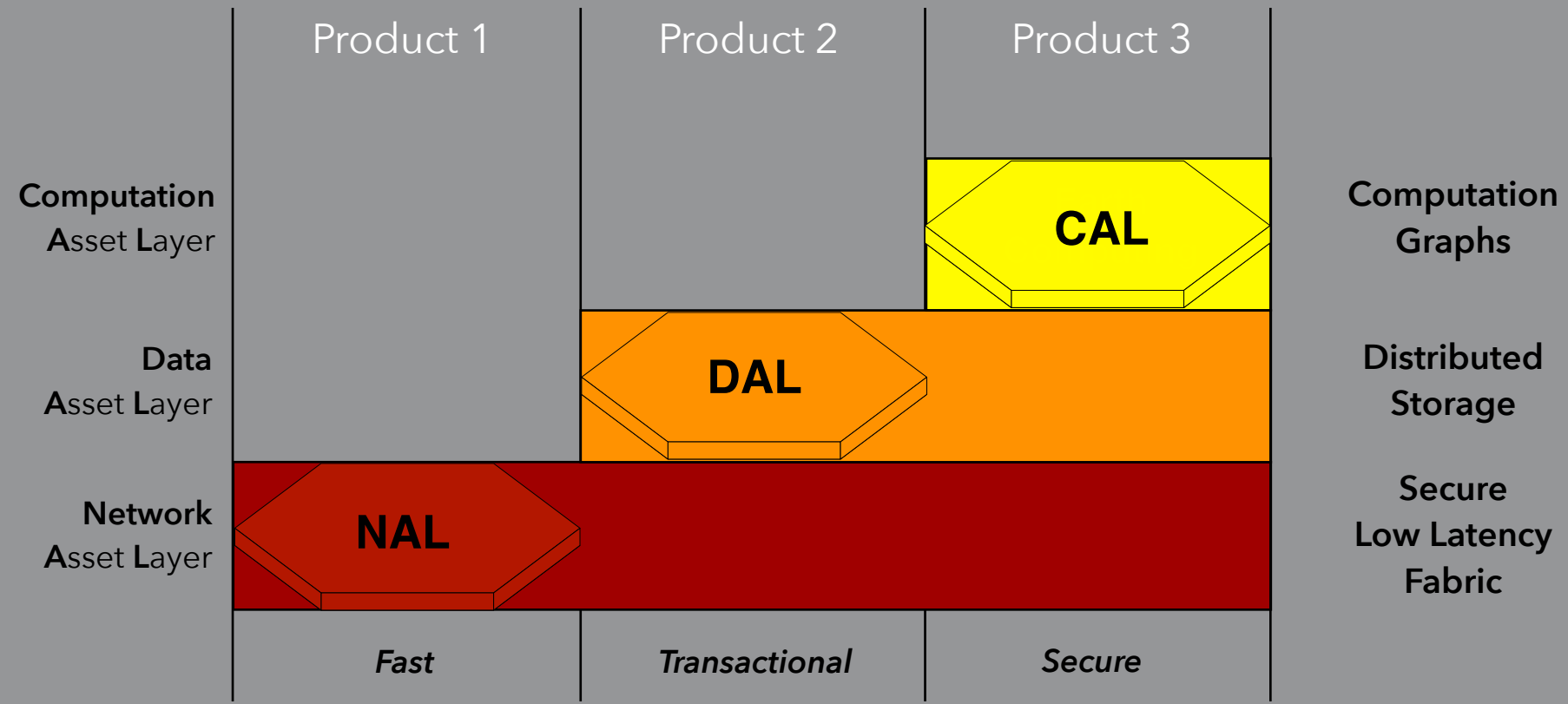
Guaranteed in the sense that both sides “know” if it failed or succeeded



- A new protocol over any physical link
- Reversible
- First class object
- No-Copying
- Composable

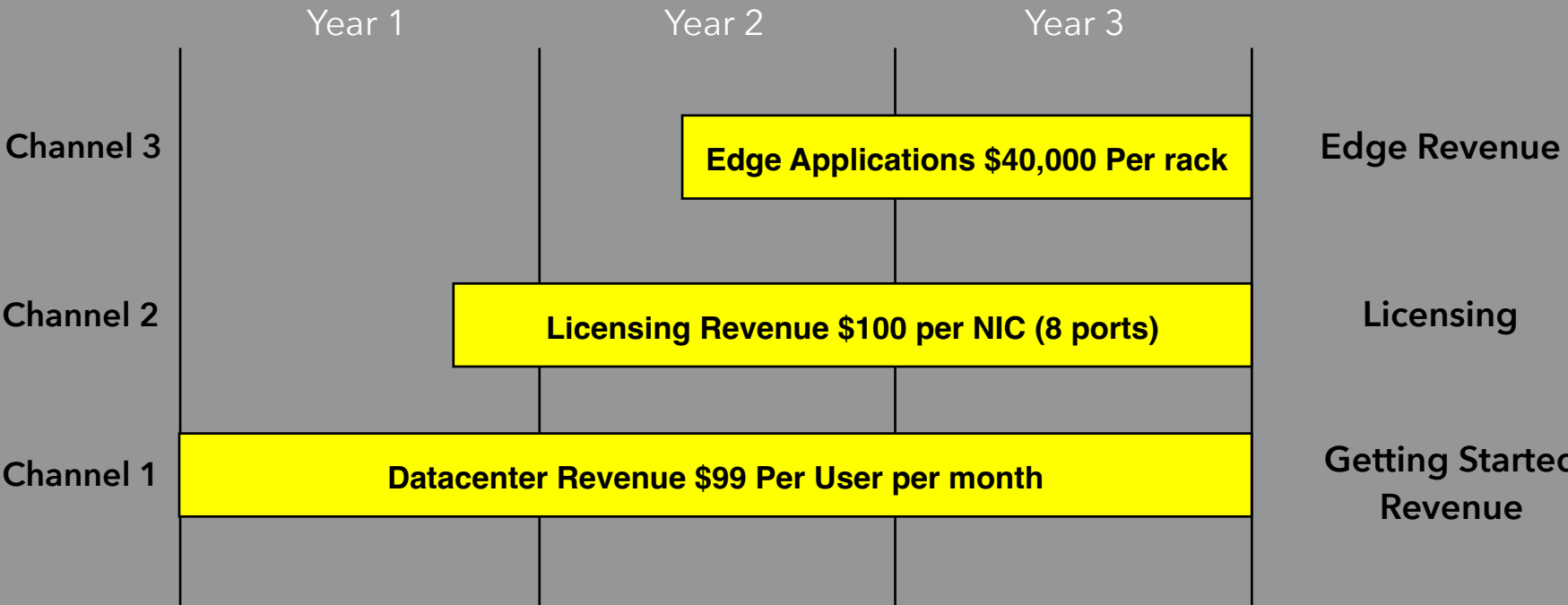
Information can't travel faster than light, but liveness appears to

Feature Roadmap



Channel Yield per Year - NAL Only

Years after Product Insertion in Market



Multi-Channel Sales Strategy

Market

A market is defined as

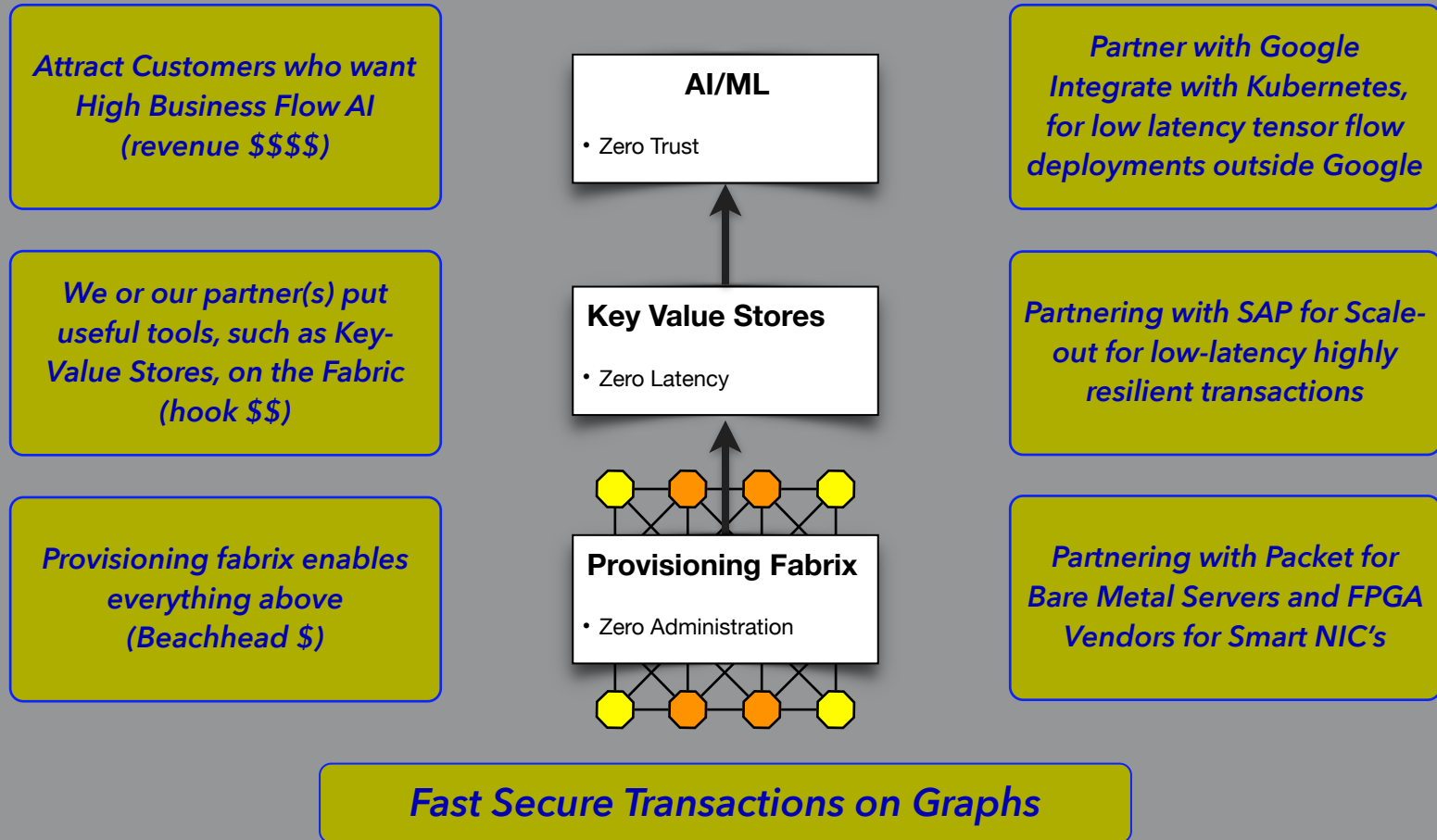
- [?] a set of actual or potential customers*
- [?] for a given set of products or services*
- [?] who have a common set of needs or wants, and*
- [?] who reference each other when making a buying decision*

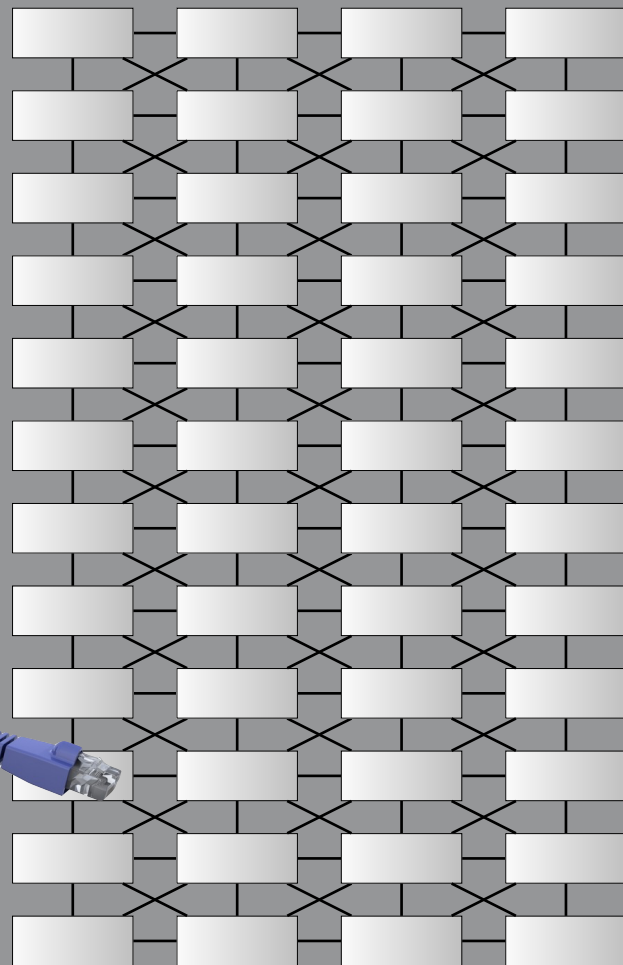
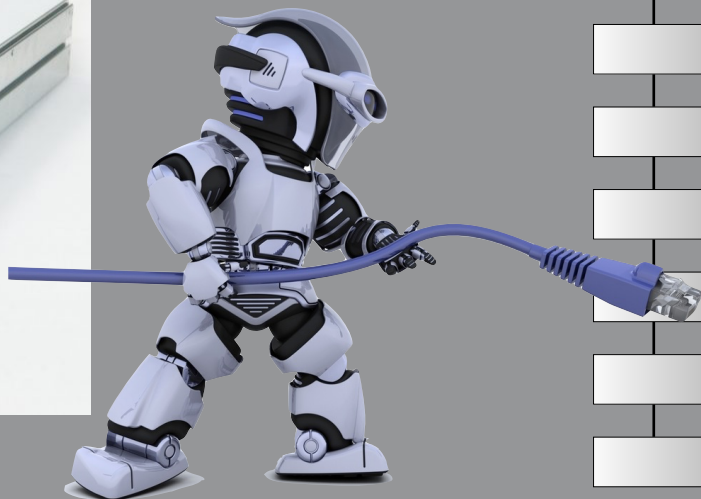
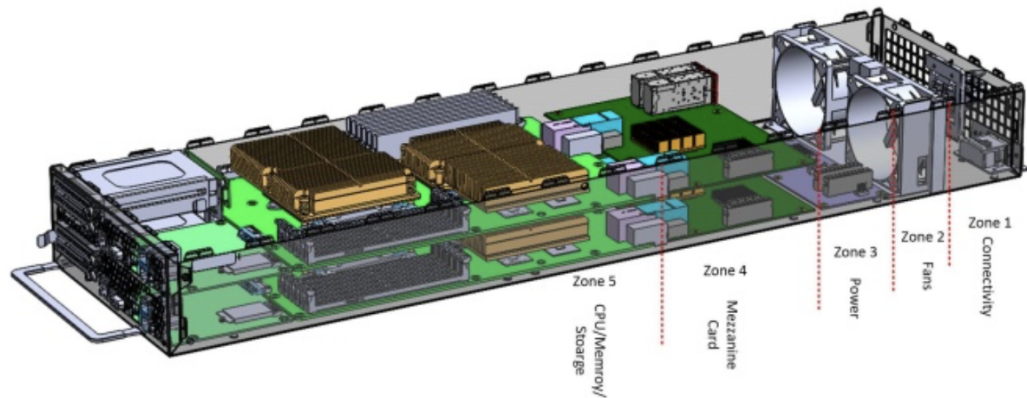
Product

Market Adoption

Early Adopters are the rare breed of visionaries "who have the insight to match an emerging technology to a strategic opportunity,... driven by a 'dream'. The core dream is a business goal, not a technology goal, and it involves taking a quantum leap forward in how business is conducted in their industry or by their customers...

Market Insertion





Customer Prospects (2020/2021)



FoundationDB



Corporate AI Tech



swisscom



COMCAST



Applied AI

NUTANIX

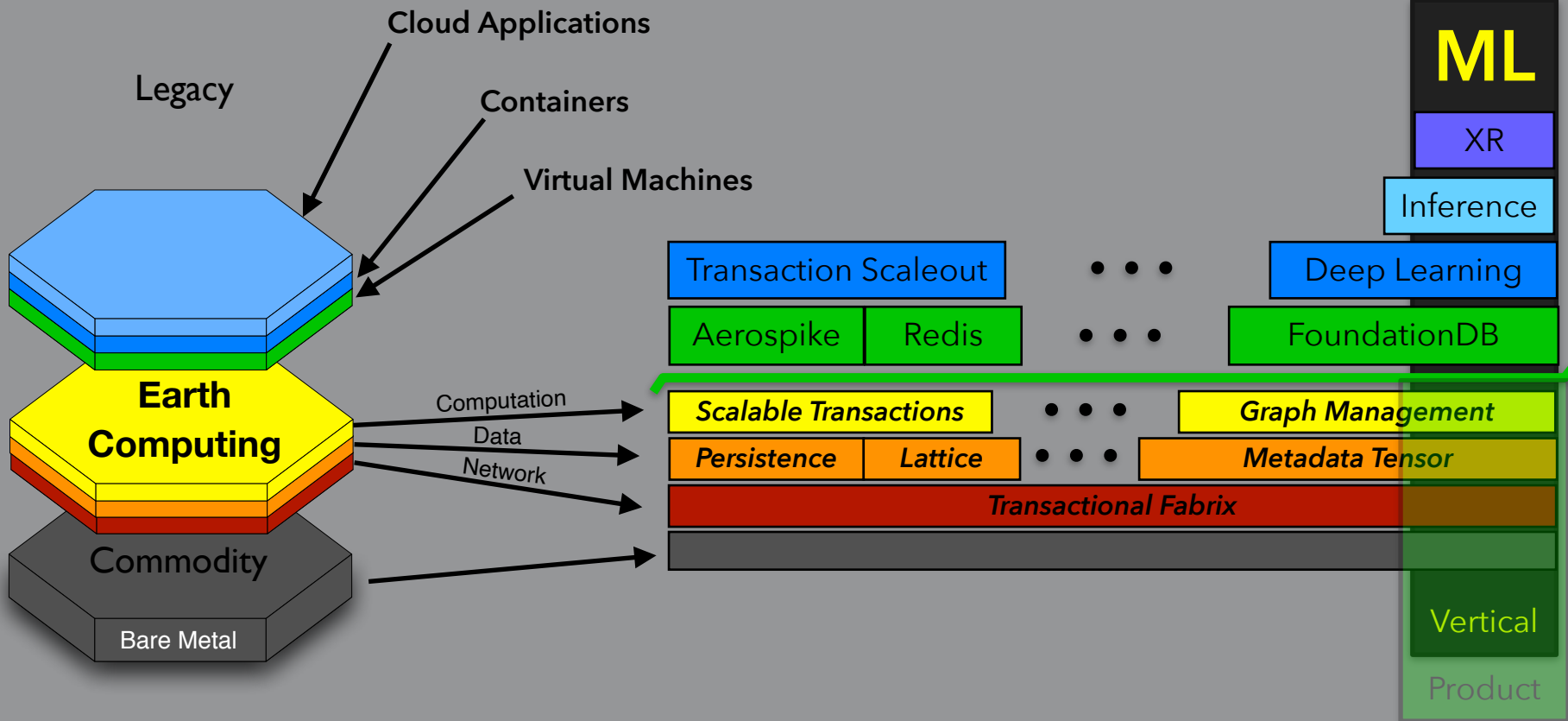
A Culture of Zero

Zero Administration

Zero Latency

Zero Trust

Vertical Applications: UltraLow Latency



Summary: Fast Secure Transactions on Graphs

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- ▶ **R**eliable **A**ddress **F**ree **E**thernet

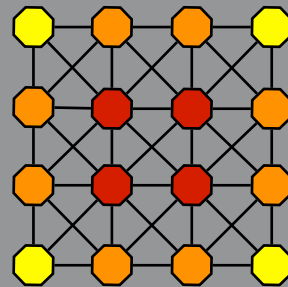
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- ▶ Intrinsic failure detection and Exactly-Once delivery

- ▶ Secret Sauce: Atomic Information Transfer (AIT)

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- ▶ Consensus Tiles for high resilience



Cellular Fabrix

Graph Infrastructure Programming

AI/ML Use Case Tree Construction

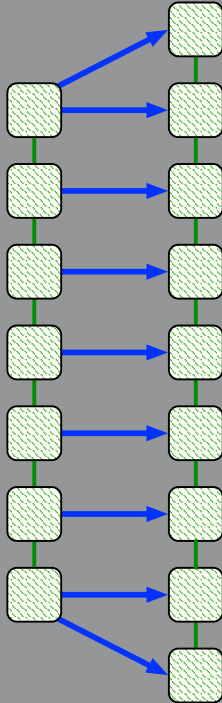
Step 1
Select
Distinguished
Cell. Root for
DNN TRAPH



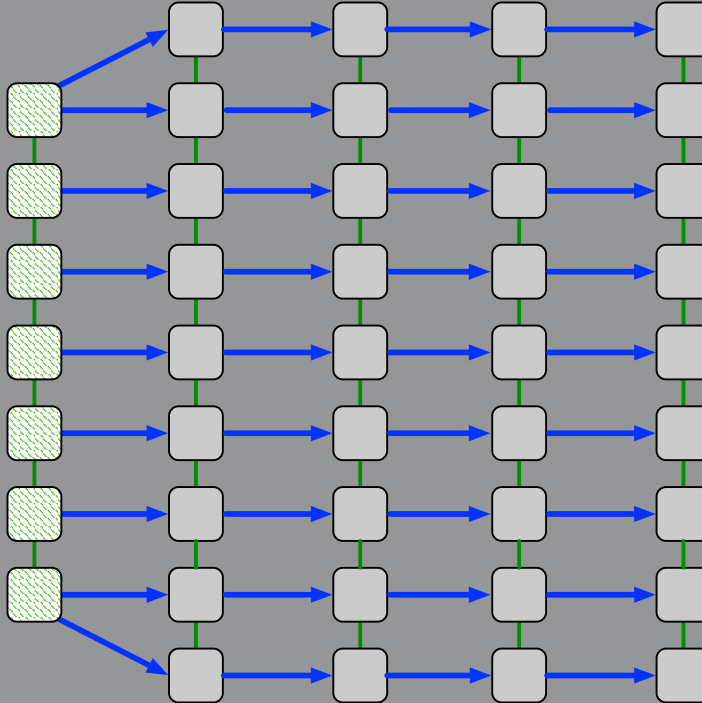
Step 2
Create input
layer with
distinguished
cell as root



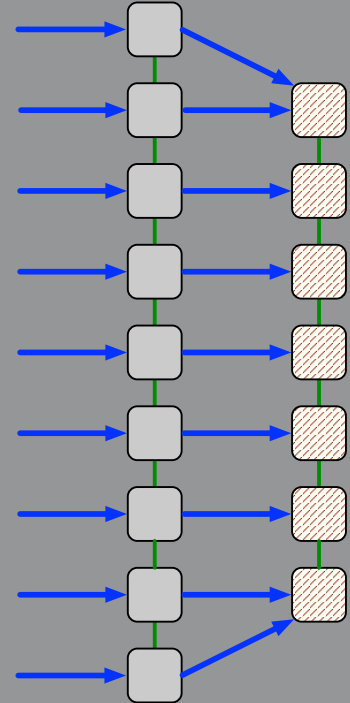
Step 3
Push GVM
equation to input
layer cells to create
first hidden layer



Step 4
Recursively create hidden
layers (parameterized by
GVM equation)

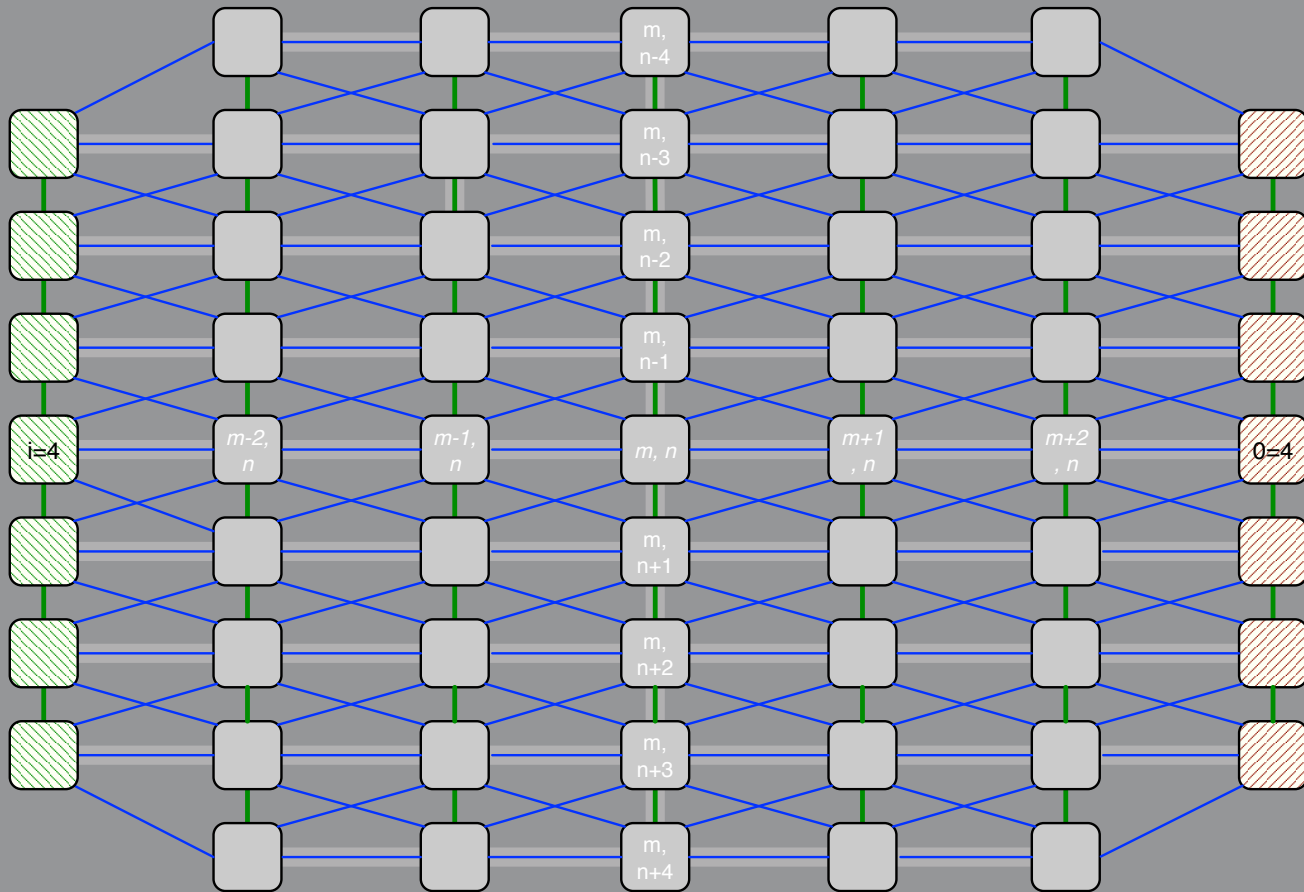


Step 5
Create output layer.
Propagate trees
back to Input Layer
for base gradient



Gradient Substrate

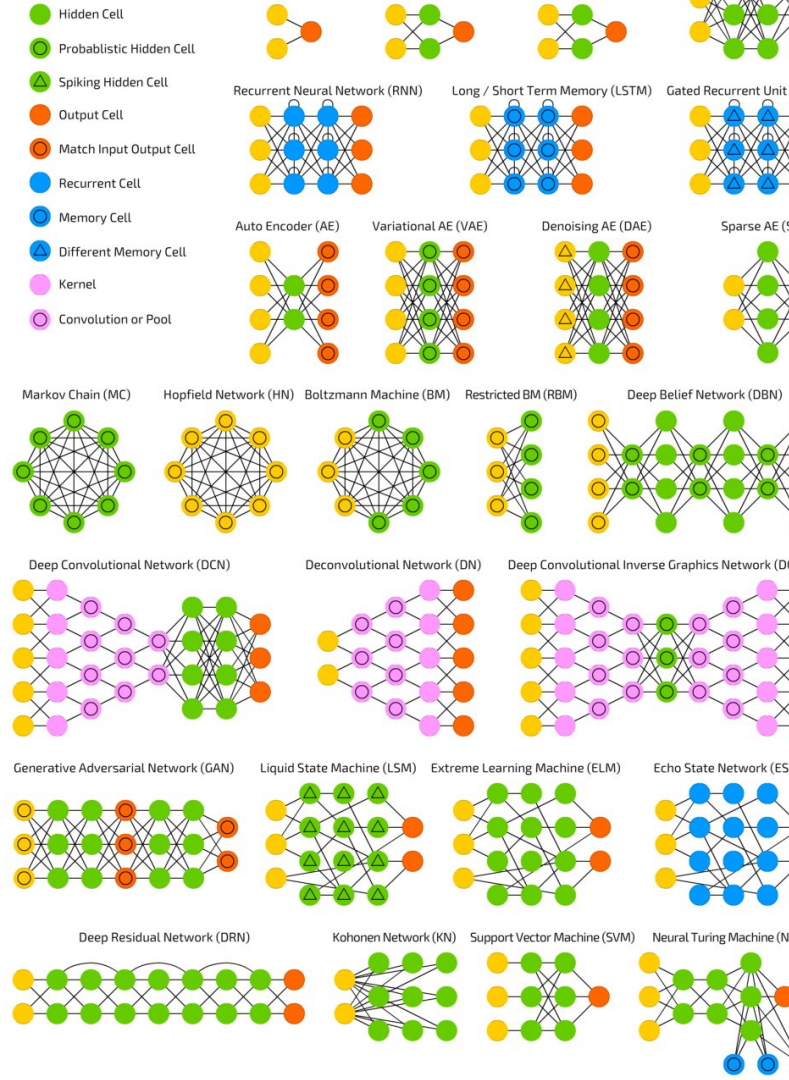
Step 6 Propagate tree roots from Hidden Layers. Establish base gradient in horizontal & vertical plane



Graph Virtual Machine

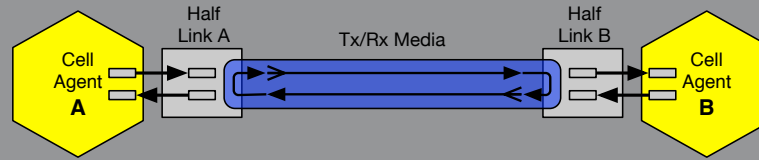
► GVM equations:

AI/ML Application Substrate:

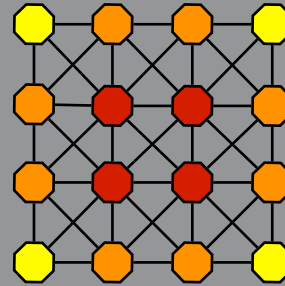


Technology

Private Transactional Link



- ▶ Transaction Performance
- ▶ Eliminate Network Latency by Bypassing Switches
- ▶ Eliminate Transaction Bottlenecks and improve transaction resilience with Quantum-like Protocols
- ▶ Eliminate Protocol Stacks, Filesystems, Double Access Penalty



Cellular Fabrix

- ▶ Eliminate OpEx with self-configuring Addresses
- ▶ Data Asset Layer with sophisticated versioning
- ▶ Integrated snapshots, backup and DR
- ▶ Misconfigured Storage buckets are a major security vulnerability

Three sub-layers of Infrastructure functionality

- The Groundplane (Physical network) comprises: (a) the Link Protocol (liveness and Atomic Information Transfer), and (b) the Stacked Tree Substrate. It can be used to provide liveness as a service, consensus as a service and resilience as a service. Legacy Compatibility is restricted to overlay paths for IP and MAC packets between endpoints
- The Local network comprises: (a) the CRDT Little Data Store, and (b) Token Transaction Layer. It can be used to provide an immutable, fast-recovery data service for configuration files and boot images
- The Virtual Layer comprises (a) stacked graphs and (b) Graph Virtual Machine (GVM). It can be used for automatic layout of microservices on the graph, and for migration of serialization foci

Opportunity: Edge

- **Datacenters**

1000ms

- **Distribution**

500ms

- **Regional**

100ms

- **Neighborhood**

50ms

s

- **Last Mile**

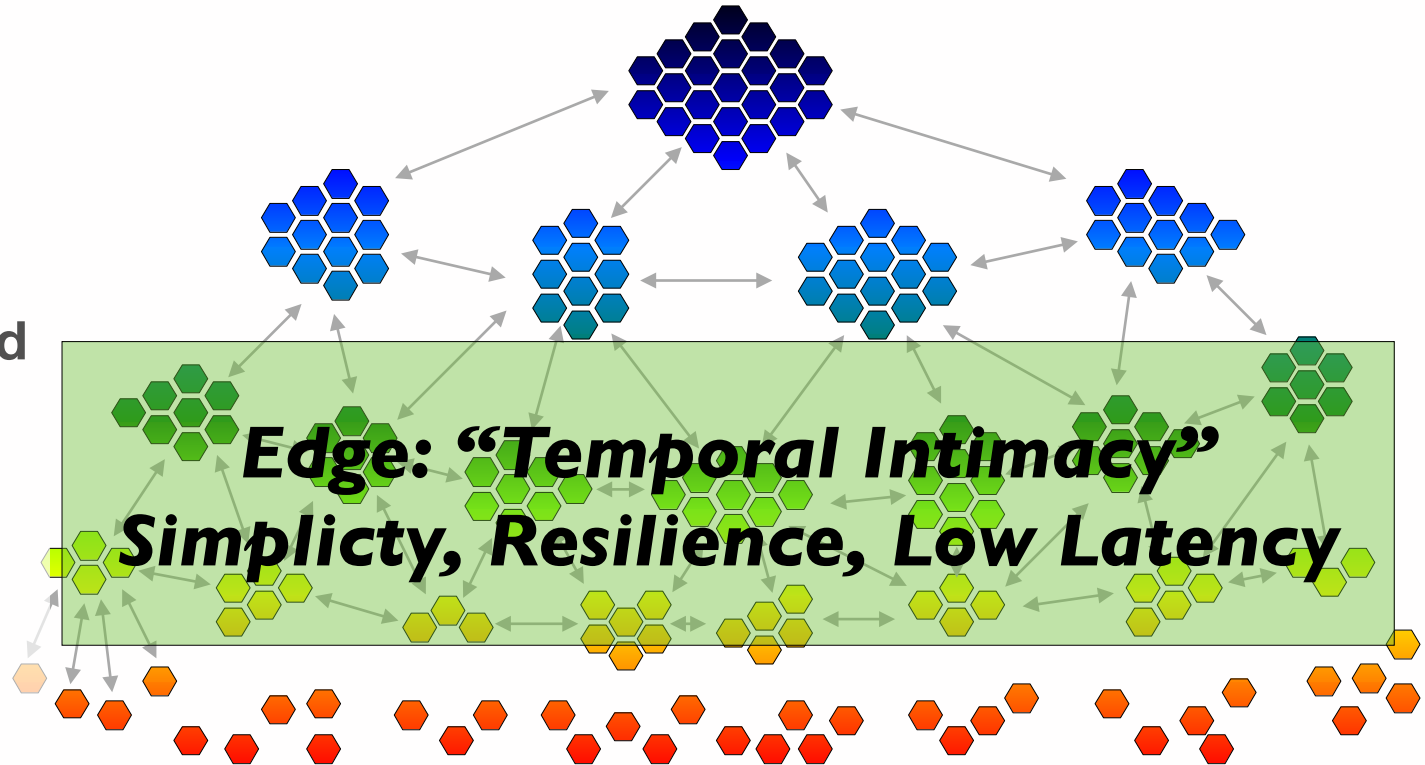
20ms

- **On Poles**

10ms

- **Devices**

1ms



Market Entry: The edge?



AT&T

verizon✓

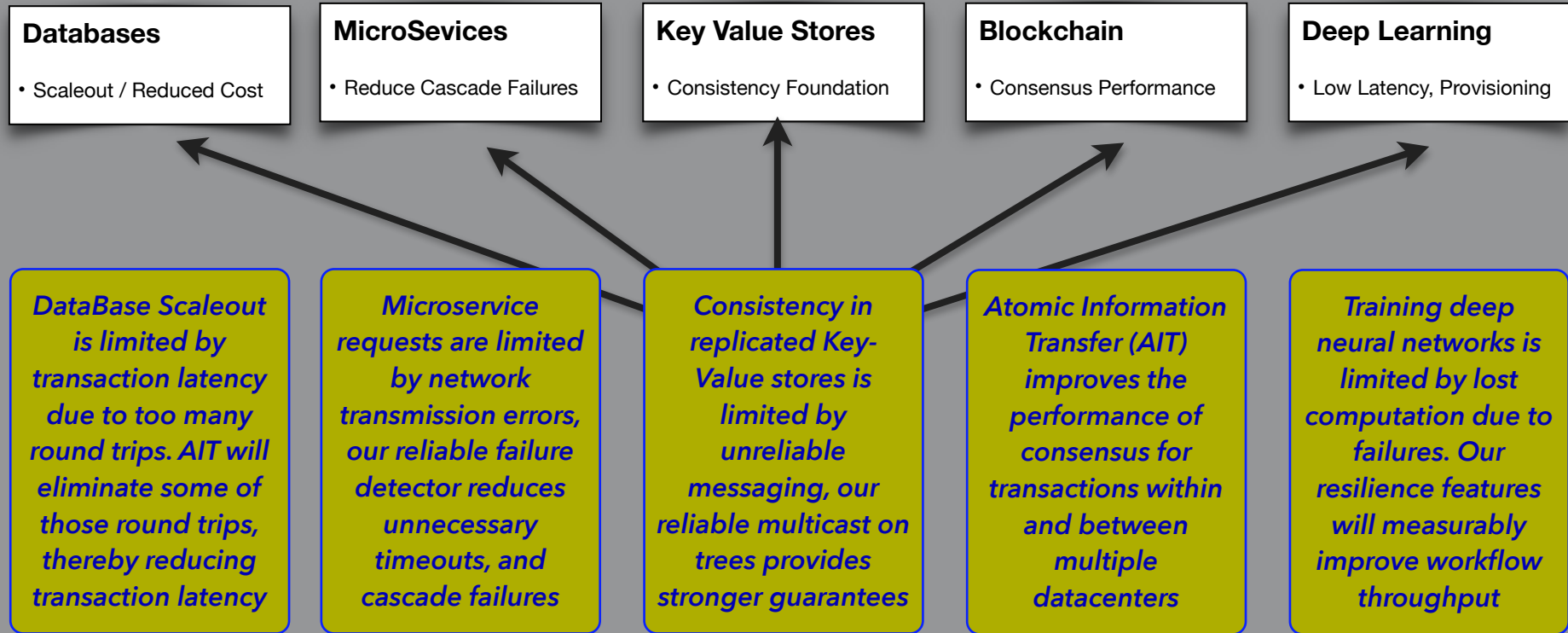
- ▶ “In order to achieve some of the latency requirements of these [5G applications?] services a service provider needs to place IT resources at the edge of the network. Especially, when looking at autonomous vehicles where you have mission critical safety requirements. When we think about the edge, we’re looking at being able to serve these low latency requirements for those [real time] applications.”

Market Entry: EC for EC



- Benefits of DAEDÆLUS to network service providers include:
 - Reduce backhaul traffic
 - Maintain quality of experience for customers
 - Reduce cost by decomposing and disaggregating access function
 - Optimize current central office infrastructure
 - Improve reliability of the network by distributing content between the edge and centralized data centers
 - Deliver innovative services not possible without edge compute, e.g. Industrial IoT autonomous vehicles, smart cities, etc

Use Cases



Chiplet Fabrix is not just Networking, they are Much Bigger

Three overlapping hexagons in a light gray color, arranged in a triangular pattern on the left side of the slide.

Questions?

Network Defined Software

Alternative Addressing: RA FE

A2A

N2N

Technical Notes: Concept Differences

[Q9] What other conceptual differences are there?

[A9] [No Token Duplication (Retries)] If a ‘sent’ packet fails in any way, we don’t retry, we ‘unsend’. The treatment is strict for packets containing tokens, it may be relaxed in the case of normal messages, where it may be assumed that legacy systems are still using TCP in the upper layers.

[A9] [Failure Notification] Unlike conventional switched networks there is no uncertainty as to whether a failure occurred or not. Failure notification is immediate, either the packet gets there, or the sender gets an immediate notification that it didn’t.

[A9] [No Retry Storms] Recovery (healing of the trees are *local*, there is no global leakage of events to cascade onto other parts of the system.

[A9] [No Gossip Amplification] With our reliable failure detector we turn a probabilistic incident (link failure) into a deterministic response (link failover) guaranteeing (locally) the containment of the response¹¹.

[A9] [Partition-Free] All links are effectively independent failure domains. Our analytic evaluations, and our simulations indicate that approximately 40% of the links must fail before the fabric partitions. Even then, both sides are informed that it has been partitioned, and can continue with our CRDT-based data structures on both sides¹².

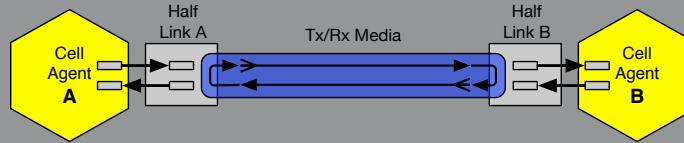
¹¹Unlike gossip protocols used at Amazon, which can lock themselves up in a mode where the protocol may not subside until

Industry Inevitability

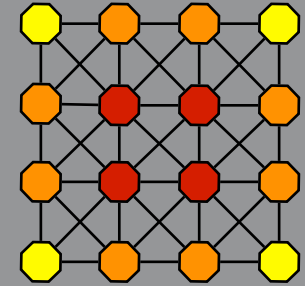
Value Proposition

Our technology enables us to provide a solution at a fraction of the cost with much higher reliability and genuine security advantages

Atomic Communication



Cellular Fabrix



- ▶ Foundational Technology (IP + earned secrets)
- ▶ World Class Engineering Team
- ▶ Incumbents Unable to Follow

ML Fabrix

- ▶ Hop Link Latency (almost always 1)
- ▶ Atomic Pebble/ Token Transfer
- ▶ Tree Addressing
- ▶ GVM for Layout
- ▶ Guaranteed In-Order-Delivery (IOD)

Training deep neural networks is limited by lost computation due to failures

*We are carving out an environment
for distributed applications that
don't need the complexity of the
general network*



DÆDÆLUS

Solid Ground Beneath the Clouds

2005 ~ 2015



NET
WORKS



ARISTA

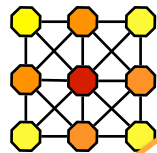
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2015 ~ 2025



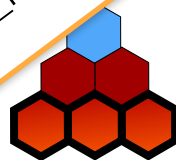
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DAE

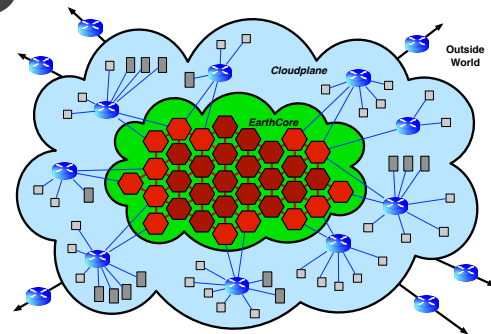


Physical

Logical



Virtual



2025 ~ 2025
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DÆDÆLUS Cellular Fabrix

Start Small

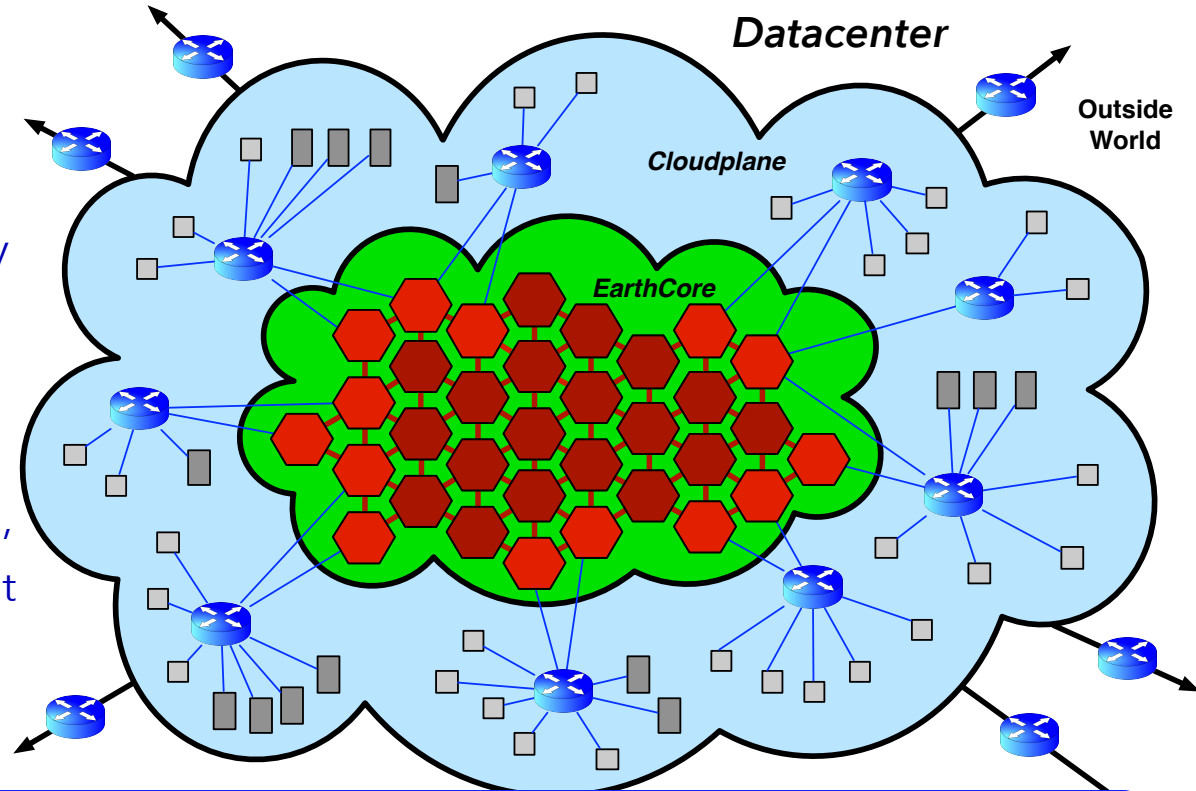
Within a rack

Cloud datacenter accessed by
untrusted legacy protocols

Earth dynamic, resilient,
programmable topologies

Core where data is immutable,
secure, protected, and resilient
to perturbations and attacks

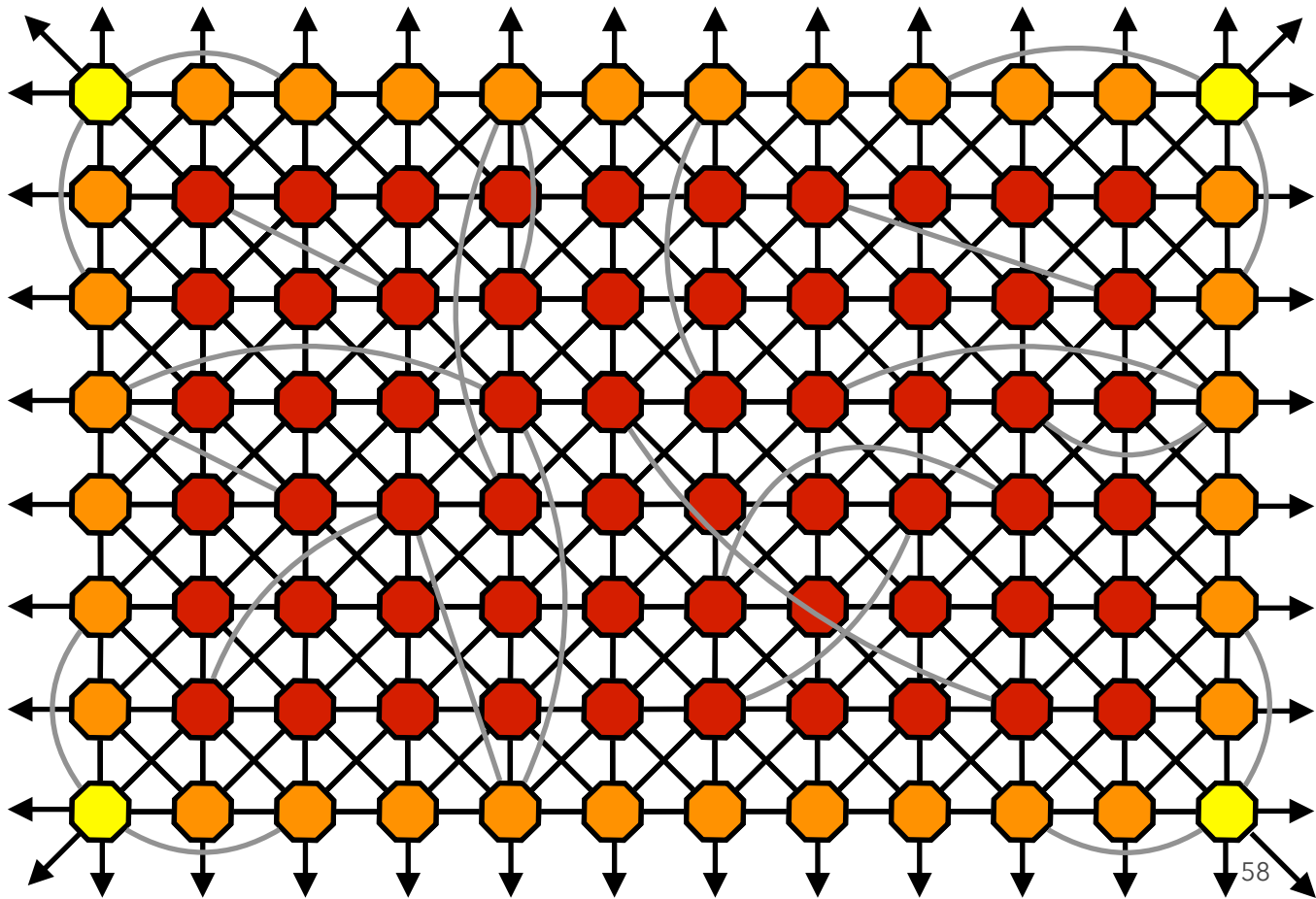
Simple, Resilient, Secure



Infrastructure Developers Wish They Had

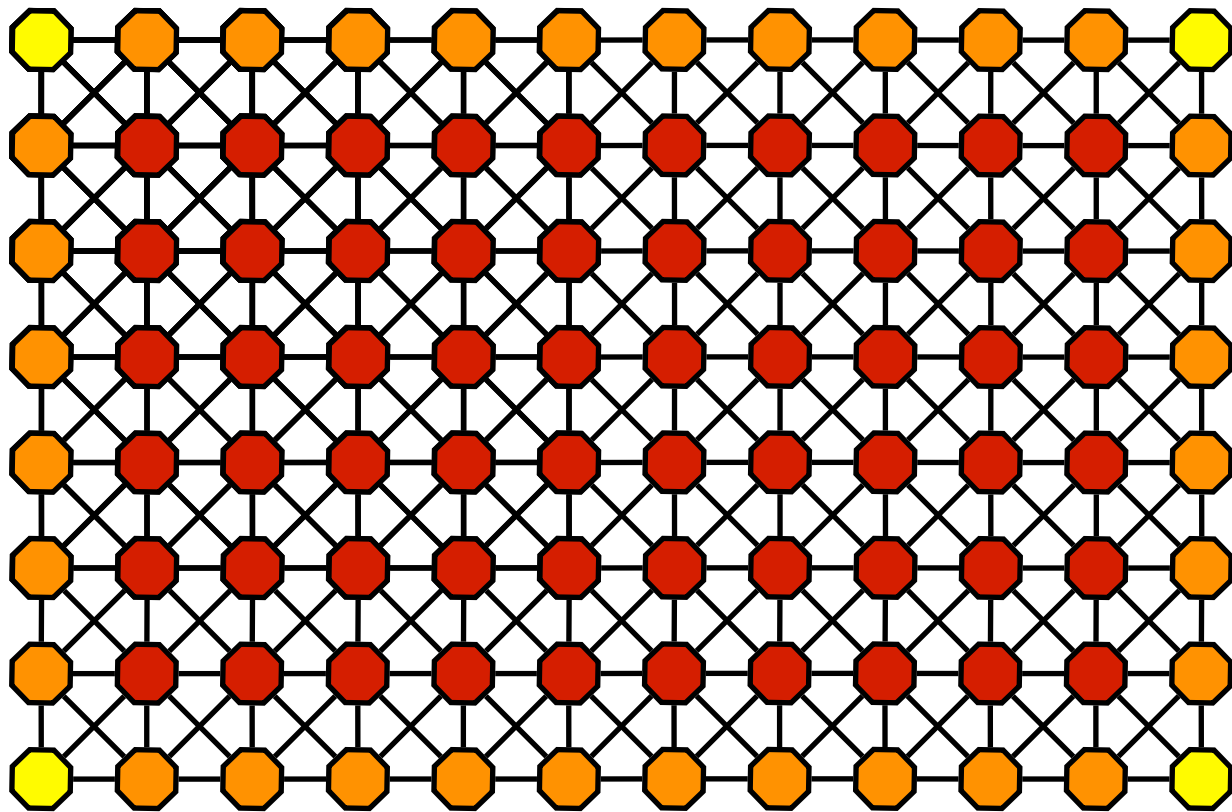
Cellular Fabrix

- ▶ Start with a Cell
- ▶ Connect only to its neighbors
- ▶ Linearly scale them
- ▶ Toss in a few Kleinberg Links to reduce average hops



Cellular Fabrix

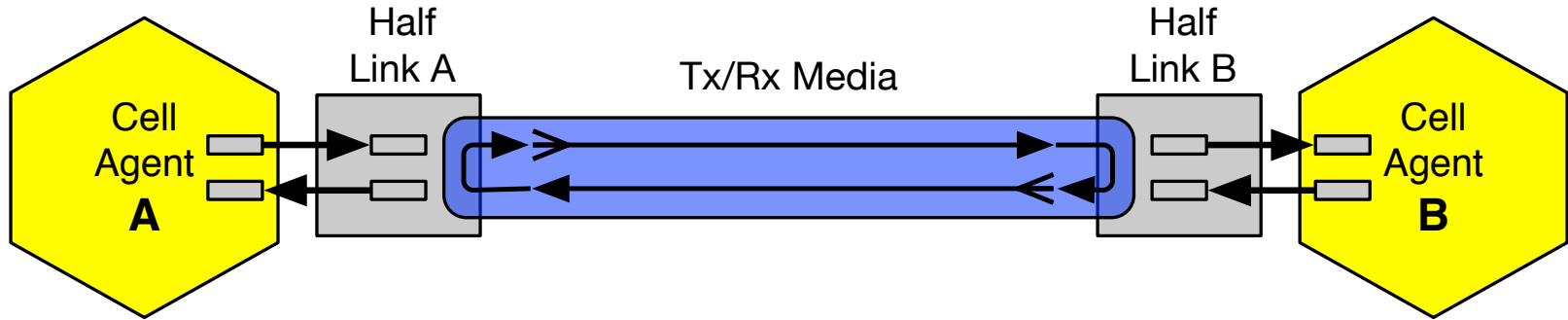
- ▶ Start with a Cell
- ▶ Connect only to its neighbors
- ▶ Linearly scale them
- ▶ Toss in a few Kleinberg Links to reduce average hops



Secret Sauce: Active Link Technology

Atomic Information Transfer (AIT)

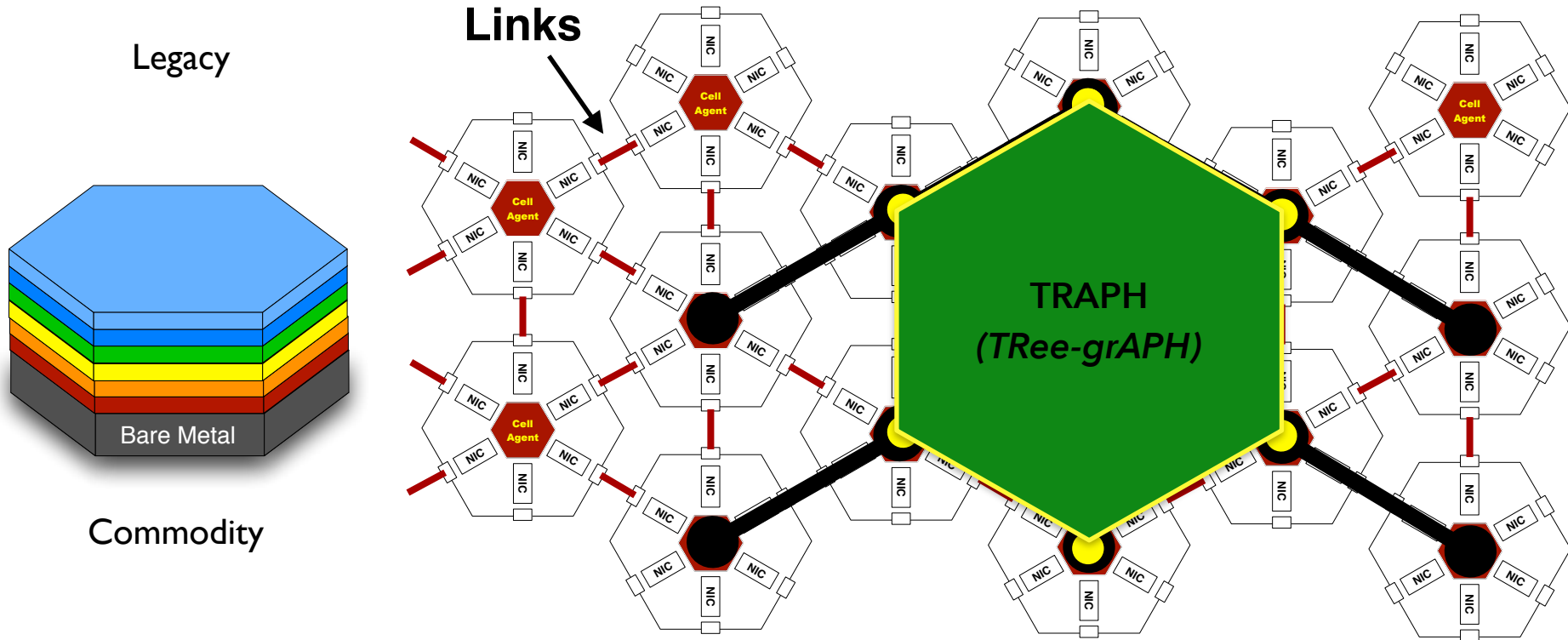
Guaranteed in the sense that both sides “know” if it failed or succeeded



- A new protocol over any physical link
- Reversible
- First class object
- No-Copying
- Composable

Information can't travel faster than light, but liveness can

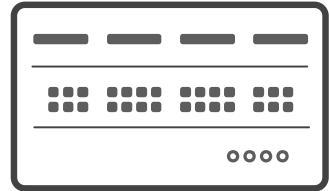
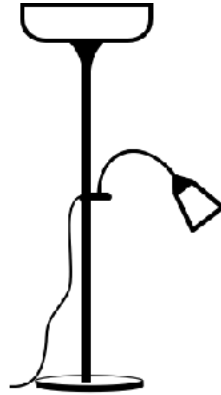
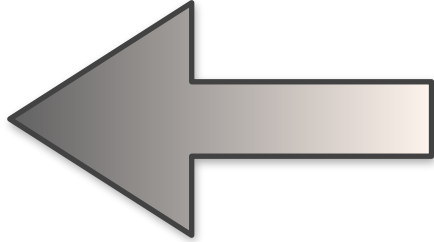
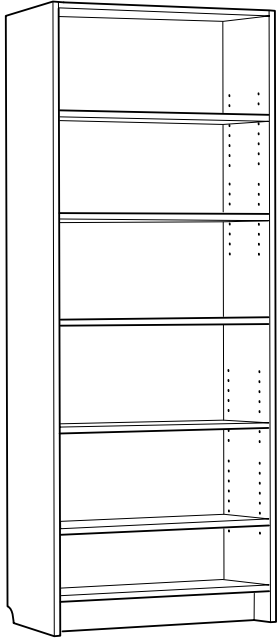
Network Defined Software



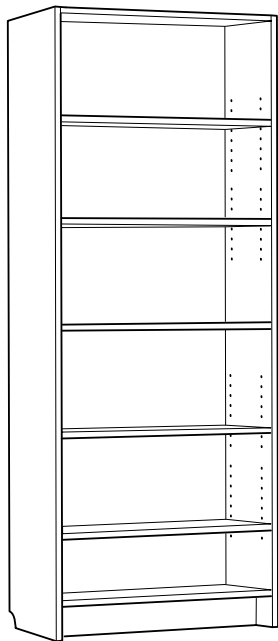
Programmable Topologies under API control for Developers

Simplicity is the only right answer

Rack *(store your computers)*



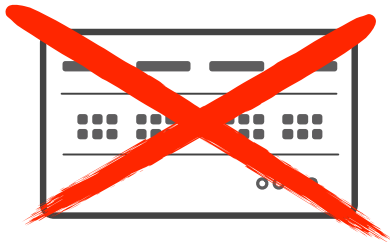
Conventional Thinking (Clos/SDN)



maximize number of hosts



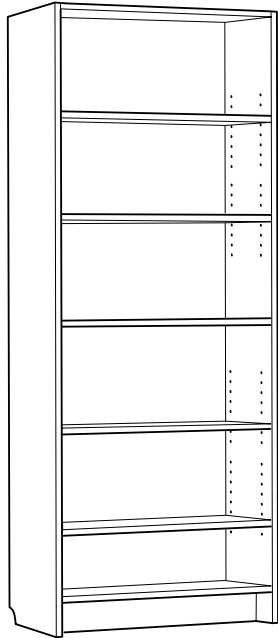
need switches for connecting to upstreams



avoid dedicated network hardware:

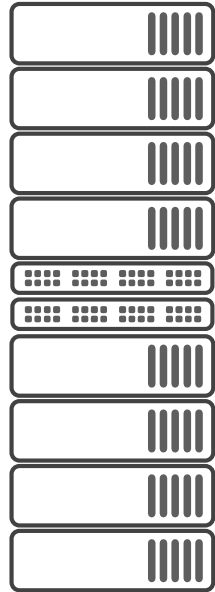
- ▶ unneeded features (routers)
- ▶ load balancer appliances
- ▶ multiple switch tiers

Problem statement

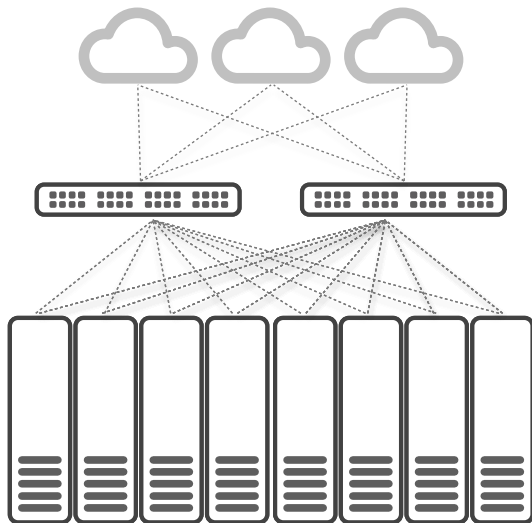


Given fixed physical footprint, how do
connect a **scale-out distributed system**
which is **simple**, **resilient** and **secure**?

Small Scale Clos Network

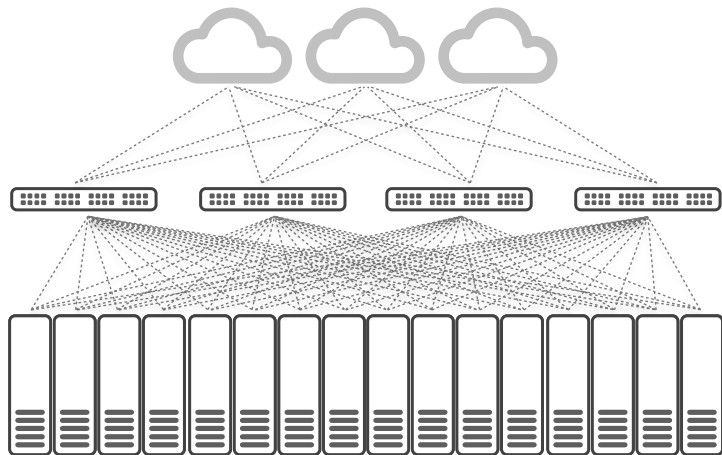


Scaling Transactions on a Clos Network



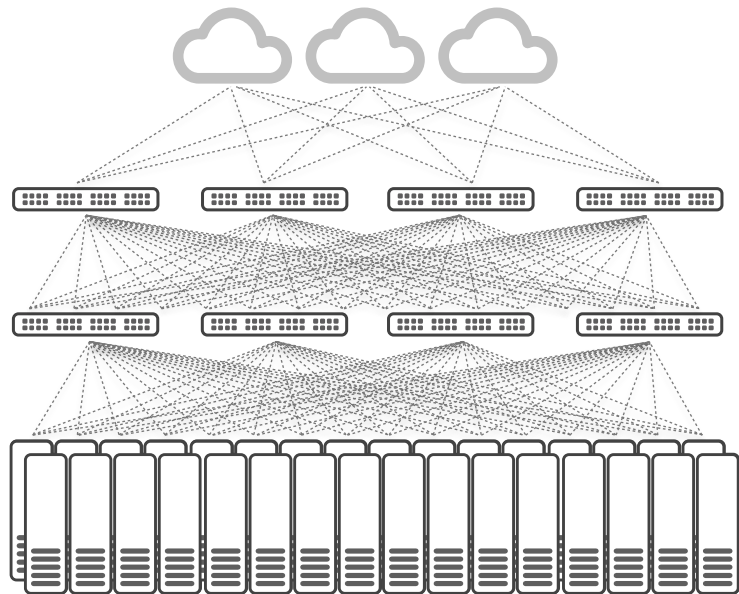
Hosts	Racks (42U each)	Bandwidth (Gbps)	Latency (ns)
8	0.5	400	400

Scaling Transactions on a Clos Network



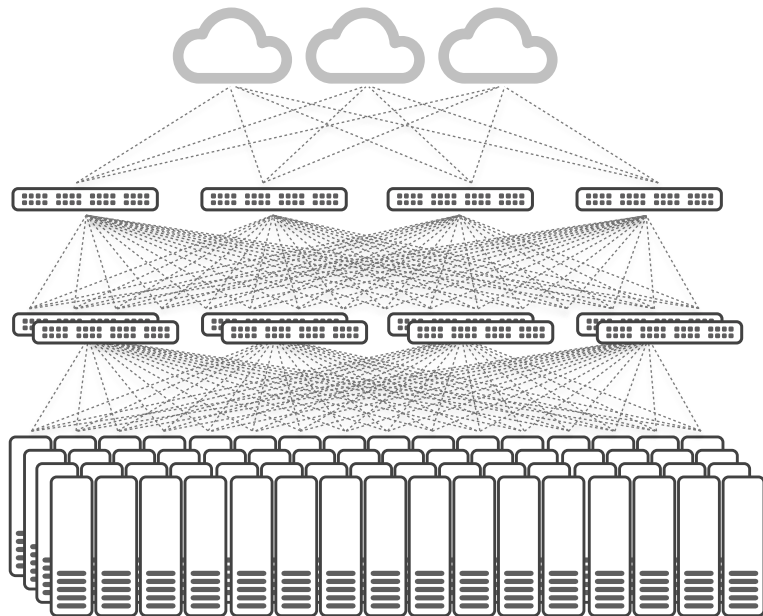
Hosts	Racks (42U each)	Bandwidth (Gbps)	Latency (ns)
8	0.5	400	400
16	1	1200	600

Scaling Transactions on a Clos Network



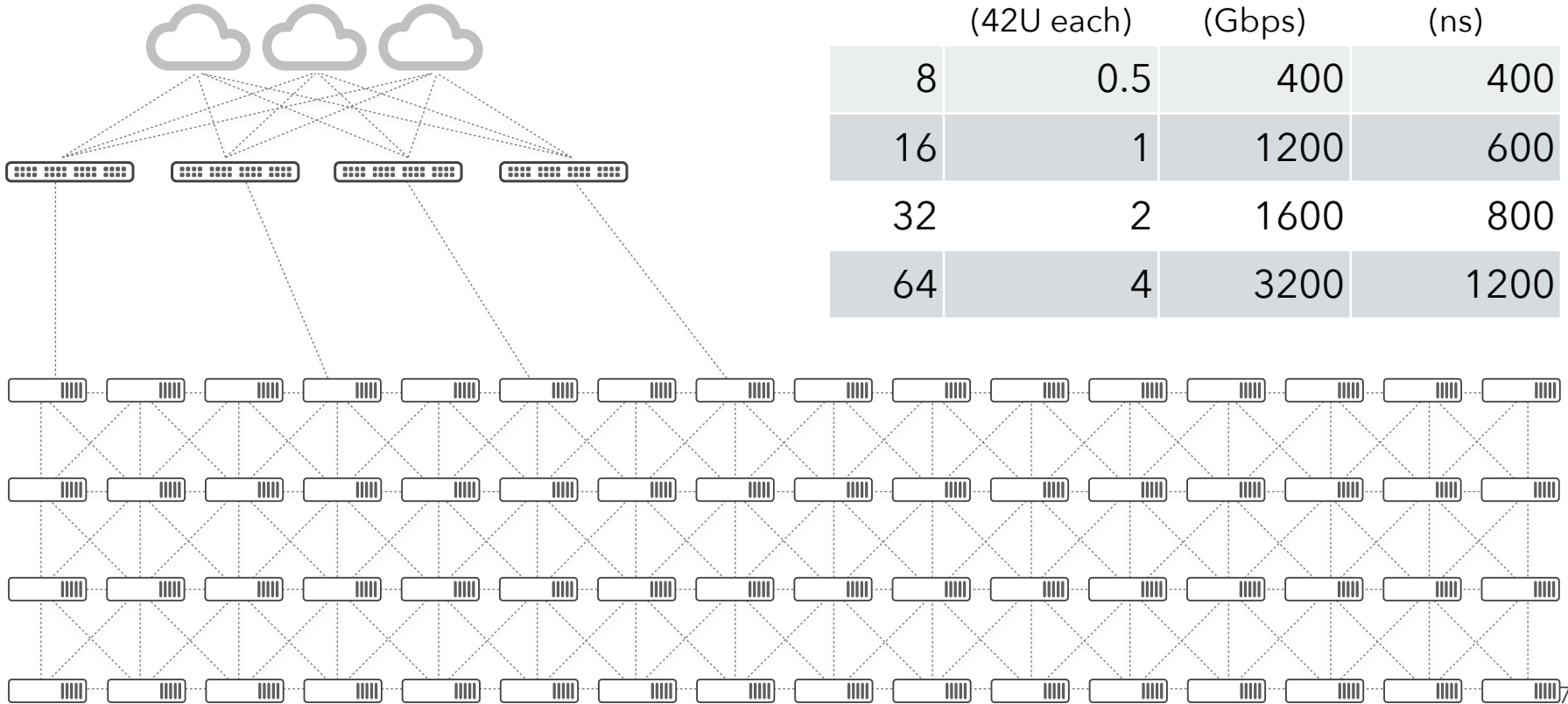
Hosts	Racks (42U each)	Bandwidth (Gbps)	Latency (ns)
8	0.5	400	400
16	1	1200	600
32	2	1600	800

Scaling Transactions on a Clos Network



Hosts	Racks (42U each)	Bandwidth (Gbps)	Latency (ns)
8	0.5	400	400
16	1	1200	600
32	2	1600	800
64	4	3200	1200

Scaling Transactions on Cellular Fabrix

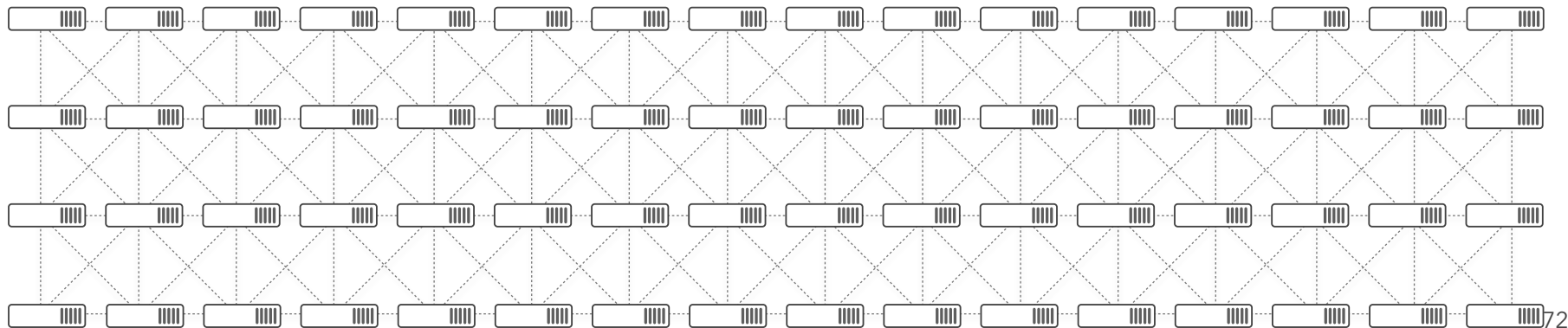


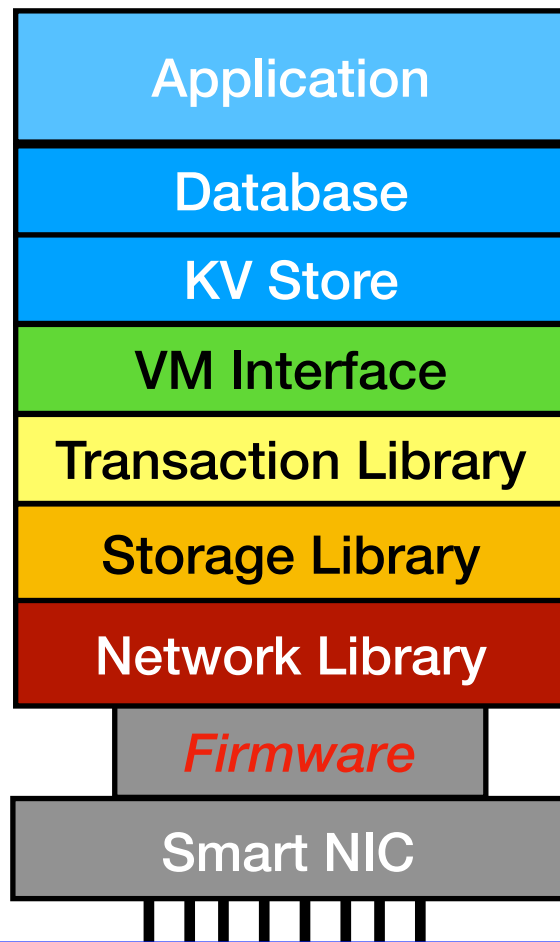
Cellular Fabrix

Cells	Racks (42U each)	Bandwidth (Gbps)	Latency (ns)
8	0.5	1600	2
16	1	4800	50
32	2	6400	100
64	4	12800	150

Clos Network

		Bandwidth (Gbps)	Latency (ns)
		400	400
		1200	600
		1600	800
		3200	1200





Legacy

**DCR KVS
Virtual Appliance**

It sits in your datacenter but doesn't interfere with it

Benefits (for network folks)

- **Simplicity:** 10x lower OpEx
- **Resilience:** 10x greater reliability
- **Security:** 10x smaller attack surface

PLUS: 10x lower minimum latency

10x available bandwidth

10x transaction rate

10x lower cost

10x higher security

10x agility

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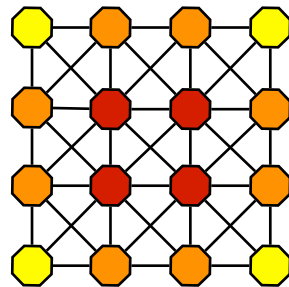


The Infrastructure Administrators Wish They Had

Benefits (for distributed system folks)

Transaction Fabrix for distributed applications

- ▶ Vastly improved resilience with Consensus tiles
 - ▶ Generalization of 2PC and Paxos
- ▶ Scale-out with much less performance loss
 - ▶ Less expensive CapEx (as many switches)
 - ▶ Less expensive OpEx (management goes away)
- ▶ Shrink TCO & Time to insight for databases



Transaction Fabrix

Innovation for the Lossless Network

Coping with Congestion



Ingress thresholds unrelated to egress buffer availability. Incast causes internal packet loss.



Priority-based Flow Control (Coarse grain). Victim flows hurt by the congested flows



Unbalanced load sharing. Elephant flow collisions block mice flows.



Unscheduled and network resource unaware many-to-one communication leads to incast packet loss

Coordinated Resources

Isolate Congestion

Spread

Schedule Appropriately

Mitigating Congestion



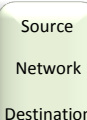
Coordinate egress with ingress decisions to avoid internal switch congestion



Adaptive congestion control. Move flows out of the way. Avoid head-of-line blocking.



Load-balance flows at higher granularity. Use congestion awareness to avoid collisions



Scheduling decision integrated the information from source, network and destination.

Innovation

Virtual Input Queues

Dynamic Virtual Lane

Load-aware Packet Spraying

Push & Pull Hybrid Scheduling

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We Empower the Developer

‣ **Business Agility**

- Speed with which to enter and exit a business opportunity
- Speed at which you can change your business model or pricing plan.
- IT Agility: The ability to scale applications, manage & confine tenants

‣ **Developer Agility**

- By empowering the developer, we improve the agility of the enterprise
- Reduce friction in developing distributed applications
- We eliminate the network complexity that app developers have to deal with

Business Agility

- **We improve IT Agility**

- But you don't care about that

- **We improve Developer Agility**

- But you don't care about that

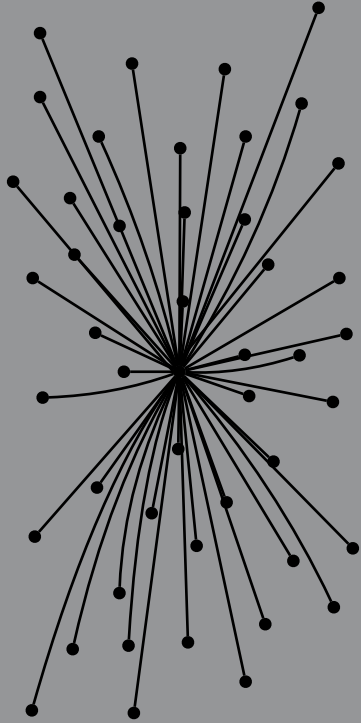
- The result is **improved Business Agility** that is all that you care about

- Enter a new business quickly

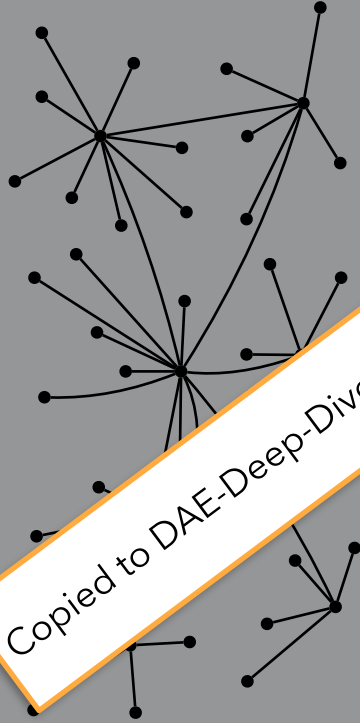
- Exist a failing business quickly

This could be the end of the story, but it's only a beginning.

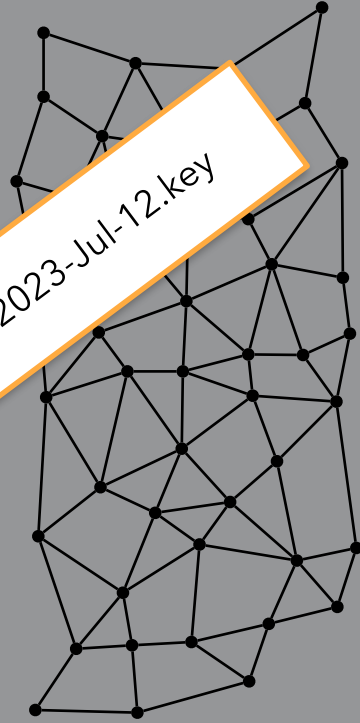
Paul Baran Classification



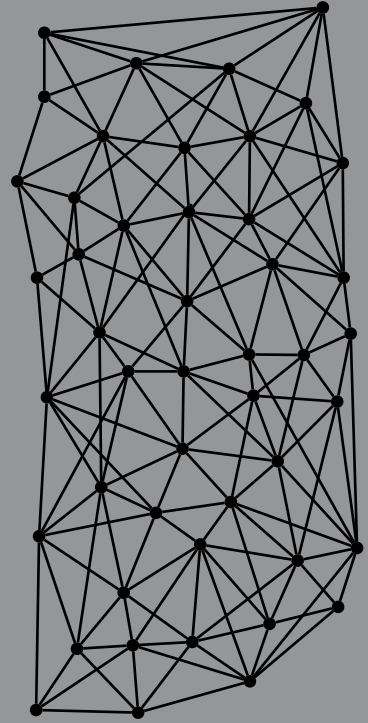
Centralized
(47 Nodes, 46 Links)



Decentralized
(47 Nodes, 47 Links)



Distributed
(47 Nodes, 98 Links)

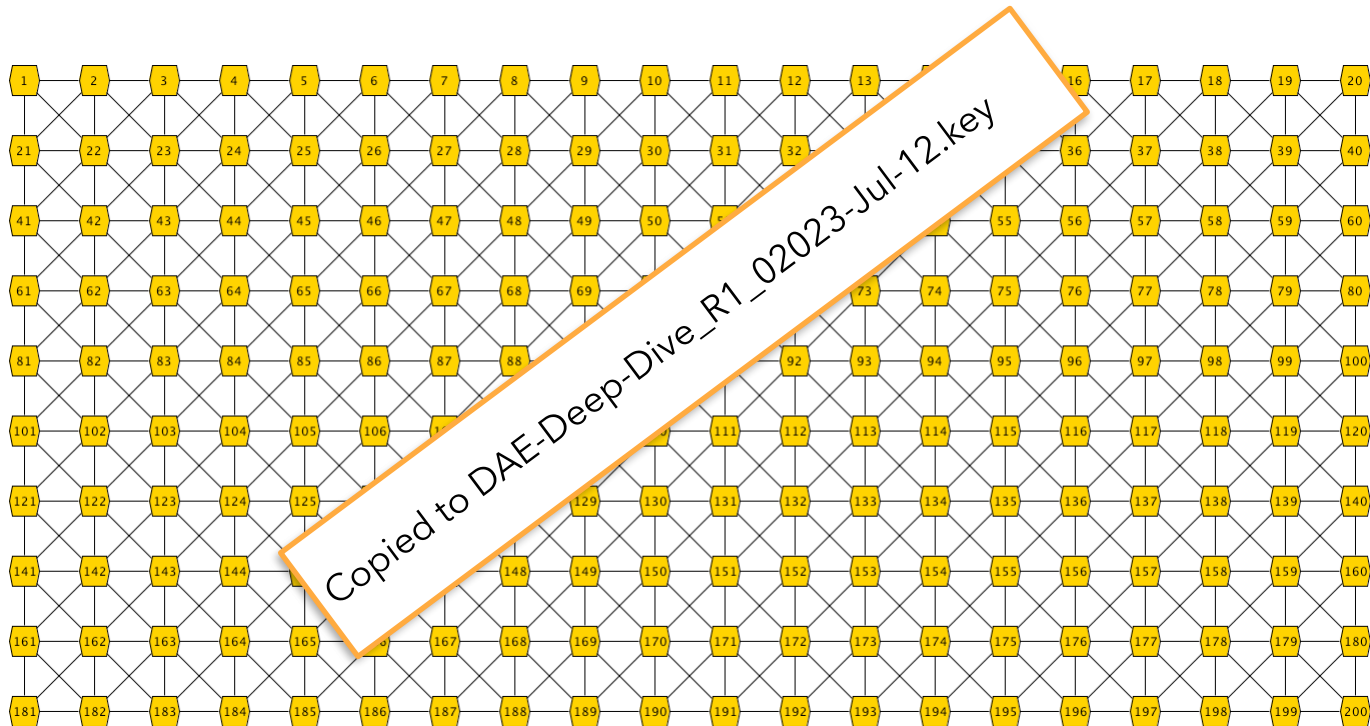


Cellular Fabrix
(47 Nodes, 220 Links)

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Regular or Irregular Lattice?

Real Infrastructures are messy

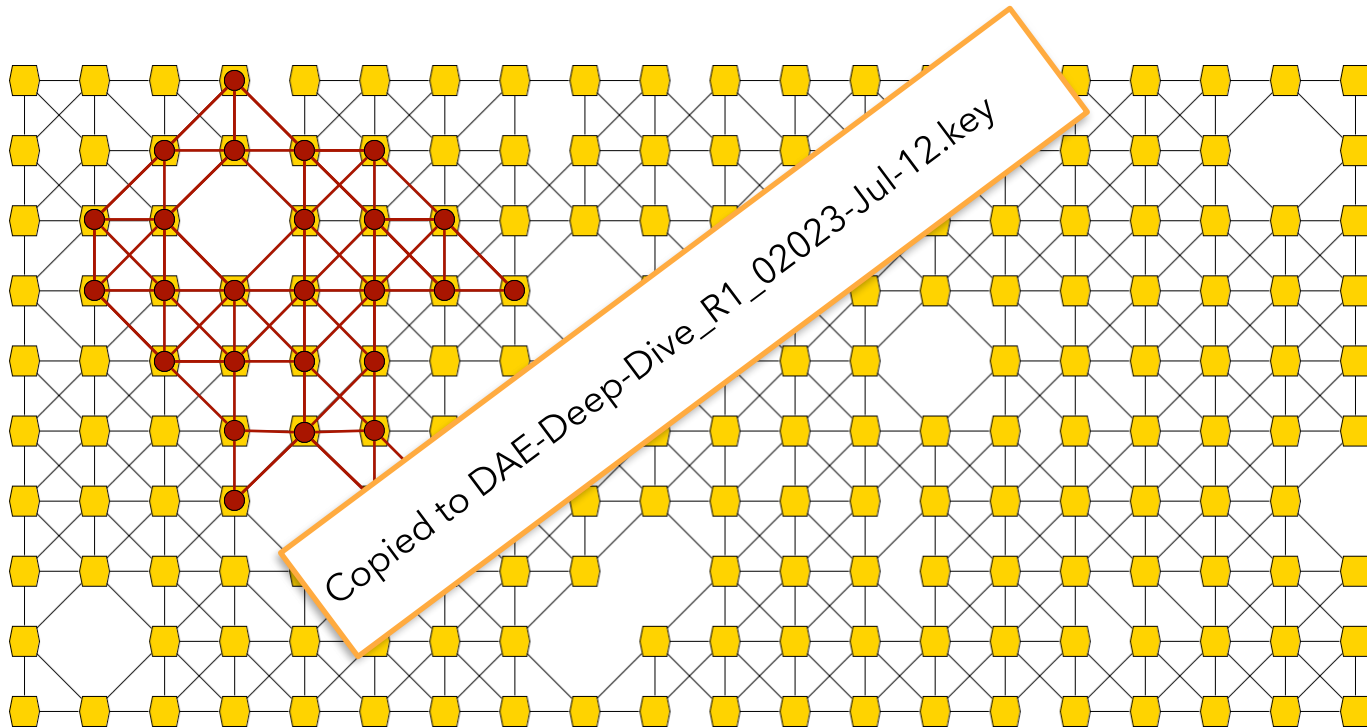


~~Absolute
Cartesian
Coordinate
Addressing~~

Neighbor to Neighbor (N2N) Lattice

Manage on a Tree

Compute on a Graph



*Relative
Address-Free
Routing*

